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Basics of Risk Management in Trade



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Steering Committee on Trade Capacity and Standards

WORKING PARTY ON REGULATORY COOPERATION
AND STANDARDIZATION POLICIES (WP.6)

Basics of Risk Management in Trade

UNITED NATIONS
Geneva, 2024

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This publication is issued in English.

United Nations publication issued by the United Nations Economic Commission for Europe.

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ECE/TRADE/485

ISBN: 978-92-1-003188-2

eISBN: 978-92-1-106554-1

Sales No.: E.24.II.E.22

Acknowledgement

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The ECE WP.6 expresses its gratitude to all those who contributed to the development of this publication.

Foreword

The COVID-19 pandemic has proven that no society is immune to risks. In a world where natural disasters increasingly imperil our future and where the triple planetary crisis of climate change, pollution and biodiversity loss creates unprecedented risks, the need for sound approaches to risk management is more urgent than ever. Moreover, as digital transformations and new technologies like artificial intelligence are changing the ways we produce, consume, and interact, questions about how to best manage related risks become more pressing – and complex.

Over the past 50 years, UNECE's Working Party 6 has worked to provide guidance on risk management in international trade. In today's globalized economy, how to best manage the risks associated to the goods and services that are traded internationally poses its own, specific set of challenges. The rise in e-commerce adds additional complexity to these questions.

Working Party 6 defines risk management as the policy measures undertaken to safeguard against hazards that stem from the quality of products and services – or lack thereof – that have the potential to cause harm or damage the health and safety of consumers and the environment. Effective risk management involves identifying potential risks, assessing their impact, and implementing appropriate measures to manage and mitigate them. The frameworks and methodologies that underpin these processes must be robust and adaptable, capable of evolving in response to new and emerging challenges.

This collection of essays builds on the wealth of expertise of renowned experts in the field of risk management, making this body of knowledge available to a larger, non-technical audience. It focusses on key aspects of the quality infrastructure for trade, including technical regulations and their enforcement, conformity assessment procedures, border controls, inspections and market surveillance. The essays also address emerging new agendas such as the transition towards a circular economy, artificial intelligence systems, and how to include gender considerations.

This publication is timely as we gather towards the Summit of the Future – aiming for a Pact for the Future – to reinvigorate the Agenda 2030 and get us back on track to achieve the SDGs. I hope the risk management tools described will help us harness the opportunities of new technologies in a way that protects human rights and leaves no one behind, strengthening trade governance for the benefit of all.



Tatiana MOLCEAN

United Nations Under-Secretary-General

Executive Secretary of the United Nations Economic Commission for Europe

Abbreviations

AI	Artificial intelligence
B/L	Bill of lading
CE	Conformité Européenne (EU conformity marking)
ECE	United Nations Economic Commission for Europe
EntreComp	European Entrepreneurship Competence Framework
EU	European Union
FTE	Full-time equivalents
GBA+	Gender-based analysis plus
GRM	Group of Experts on Risk Management in Regulatory Systems (under WP.6)
IB	Inspection body
ISO	International Organization for Standardization
ITC	International Trade Centre
LoC	Likelihood of compliance
KSA	Knowledge, skills and attitudes
MSA	Market surveillance authority
OECD	Organisation for Economic Co-operation and Development
RBI	Risk-based inspection
SDG	Sustainable Development Goals (of the UN Agenda 2030)
TBT	Technical Barriers to Trade Agreement (of the WTO)
UN	United Nations
UN/CEFACT	United Nations Centre for Trade Facilitation and Electronic Business
WCO	World Customs Organization
WP.6	Working Party on Regulatory Cooperation and Standardization Policies
WTO	World Trade Organization

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Introduction

The clearance of goods to enter a market is often considered a balance between facilitating and controlling. The latter is checking that the goods are compliant with all relevant rules and regulations, corresponding to legitimate regulatory objectives. This may include consumer health and safety, safeguarding the environment, compliance with social norms or aiming to achieve overarching objectives like the United Nation's Sustainable Development Goals (SDGs).

Risk management is the key method to achieve this balance. Unless absolutely every single shipment is to be physically verified, agencies will need to establish criteria to choose which shipments to control and which shipments to not to control. This will be through risk management.

Core concepts of risk management in regulatory systems

The first chapter is on regulation, explaining how and why Governments regulate. It recognizes the linkages of risk management with traders and how regulations impact their operations. It further provides some examples from the European Union (EU) on regulatory tools. Reforms to regulation which are now occurring widely are normally done with a view to respond to perceived risks.

This is followed by a chapter on market surveillance which clearly explains that it would be impossible to control every shipment and it is necessary to target specific products, sectors and markets which would have the highest impact, using a risk-based selection method. This chapter outlines some of the criteria used in a risk-based approach and the methodologies of control and inspection.



The third chapter explains how customs agencies utilize risk profiling. Customs is often the agency at the border that performs controls for all relevant agencies. The chapter explains the different sources of data used for and the six phases required for risk profiling. It underlines that intelligence-driven risk indicators and relevant criteria is solely driven by the collection of data.

Conformity assessment plays a key role in ensuring that products placed on the market are safe. This chapter establishes the importance of this role and follows with the various methods of conformity assessment, underlining that the method is based on an assessment of the risks that may arise as a result of non-compliance. It also provides the example of the EU conformity marking "CE."

There is not one sole Government agency which handles regulations linked to international trade; this is implied in the chapters above. In order to avoid over regulation or conflicting regulations, there is a need for a certain level of coordination. This could best be achieved as described in the next chapter through integrated risk management which incorporates a national strategy and a multi-layered roadmap. It also emphasized the importance of involving key stakeholders, including those from the private sector.

This section closes with a detailed analysis of business risks which Government agencies would take into consideration in their risk analysis. It outlines the basis of business risks, then explains methodologies to address these (from a Government perspective). It provides detailed examples from the United Kingdom of Great Britain and Northern Ireland and Armenia.

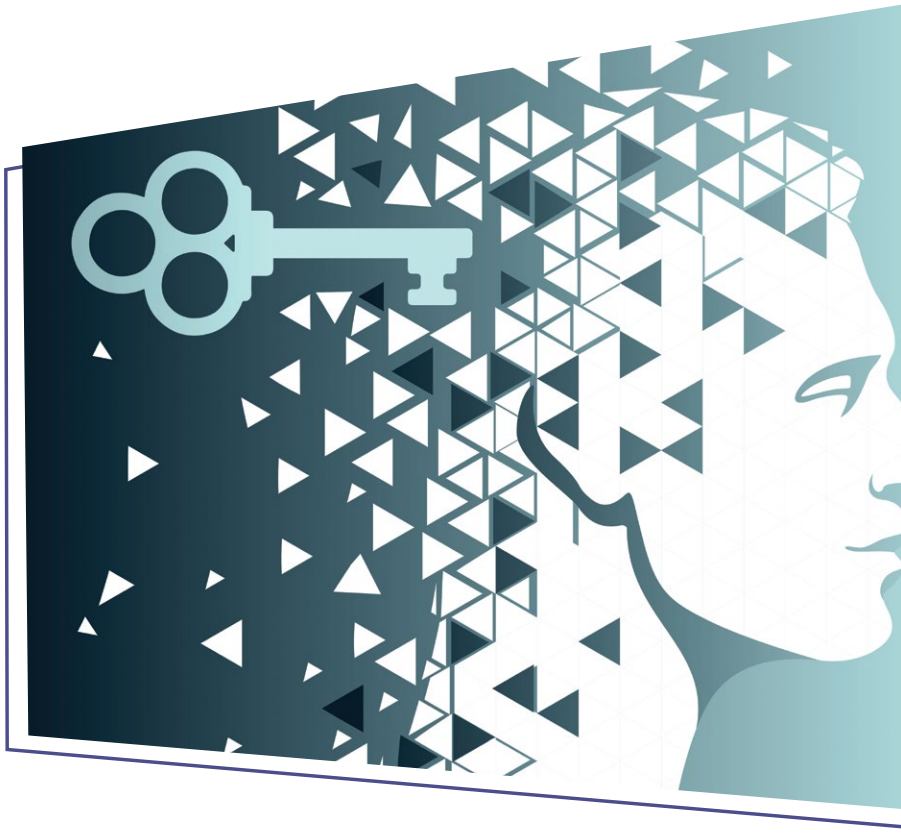
Risk management in businesses

The next section considers risk from a private-sector perspective. It is important to underline that commercial operators have a financial interest in ensuring that their goods are compliant and that they avoid unnecessary security threats; their main interest is to trade their goods and reap the financial benefits – loss due to non-compliance or security threats would result in loss of revenue.

The first chapter in this section looks at how private sector companies integrate risk management into their internal processes. The types of risks outlined in this section may differ from the types of risks that Government would consider. However, this chapter demonstrates how technology and notably advanced compliance systems can assist in addressing these risks which could provide inspiration for Government agencies.

The following chapter turns the view of risk management to the supply chain notably international transportation. It demonstrates how electronic data exchange can contribute to data collection, increased transparency and minimizing errors, delays and disputes. It recognizes the key importance of compliance with regulations and also addresses security threats. The approach is again towards digitalization and how this can help to address potential risks. The chapter does also underline the importance of stable standards like those developed by the United Nations Centre for Trade Facilitation and Electronic Business (UN/CEFACT) to achieve the corresponding electronic solutions.

This section ends with a chapter on risk competence (for private or public sector). It explains that simply applying risk management tools is not sufficient for effective risk management; it is necessary to have a certain level of analysis and develop a strategy for the organization. The chapter develops elements of risk competence and how to acquire these skills.



Emerging issues

The next section provides a forward-looking view of themes related to risk management. The first chapter on circular economy stresses that rational regulations are often better accepted by traders. It provides three case studies which highlight how a risk-based regulation helps to move towards a more circular economy. It concludes, remarking that some States may have capacity shortfalls, making it difficult to ensure proper monitoring and successful implementation.

The next chapter dives into regulations around artificial intelligence (AI) and the notion of residual risk. It takes the *EU AI Act* as a starting point. It reminds the important role of standards in demonstrating compliance and that there may not yet be stable standards on AI that can be used. Furthermore, the difficulty to analyze all of the potential functionalities and outcomes of an AI system makes testing compliance difficult. The chapter proposes how to evaluate residual risk for products with AI systems.

This section concludes with gender considerations for risk management. It underlines that hidden bias, including through scientific research and international mechanism, can perpetuate a detrimental gender bias. It explains that standards are not currently protecting women as well as they protect men. It provides a methodology used in Canada which could be used to help inform a risk management process that goes beyond gender-responsiveness.

Case study

The final section provides a case study from New Zealand which demonstrates how an electronic system has helped to assess the risk of non compliance. It outlines the types of interventions and the probability factors and technical factors built into the risk engine. It does underline the need for continuous review and update of the system.

Part 1:

Core concepts of risk management in regulatory systems

Risk management in **regulation and regulatory reform**

When goods, services and/or money are exchanged, this exchange is subject to a wide variety of rules and conditions. Some of these rules and conditions are simply agreed by the parties, who intend to exchange goods, services and/or money. Some are imposed by service providers, who enable the intended exchange, like freight conditions (size, weight, time) or price. A third set of rules and conditions is established by jurisdictions; this third group of relevant rules and conditions can be summarized as regulation. This chapter focusses on the nature of regulation and its links to risk management.

In a nutshell: Governments regulate to manage perceived risks.

Overview: regulation and regulatory reform

Governments have in general four options to act: taxing, spending, regulating and – as a soft skill – communicating. There are no different means, when Governments aim to influence the behavior of people or entities under their jurisdiction. In some legal contexts taxing and spending are even understood as specific types of regulating as both taxing and spending require Government's decisions, which affect third parties. Sometimes communication is understood as an alternative means of soft regulation.

Administrative implementation and enforcement may also be included in this general categorization. Any function of Government needs to be financed by spendings. Money provided for these functions needs to be collected – usually by taxing and similar forms of public fund raising. Last but not least, before anything is subject to implementation and enforcement, it must be defined with an aspiration to be noticed, in other words to be regulated.

Regulating might, thus, be understood as the core lever of Government action. This applies to all Governments of all types and at all geographical levels. It is also applicable to international and supranational organizations. Even treaties between jurisdictions or groups of jurisdictions, e.g., on trade issues, utilize – implicitly or explicitly – regulatory tools.

The question of what is regulated and how the quality of regulation is ensured is of utmost relevance for those concerned and for the Government itself. Those concerned in the present context are those who want to exchange goods, services and/or money; their primary objective is the commercial transaction. Any external influence, regardless of who exerts this influence, will make trading more complicated, more burdensome and more susceptible to malfunction. In other words, any external influence raises risks.

In a very similar way, the primary objective of Governments is to ensure that their interests are realized. Governments therefore aim to regulate effectively. In many cases they also care for the efficiency of their regulation, as they assume, that efficient regulation may also be more effective. All parties have a fair interest in accessible, predictable, easy to follow, and clear conditions. From a trading partner's point of view,





ideally these rules and conditions are completely known before decisions are made.

In the context of risk management in trade, it is also instructive to understand why Governments take action; either they intend to achieve certain conditions in their area of competence, or they are eager to avoid certain conditions. Governments regulate to manage perceived risks. In both scenarios, decisions of what and how to act are expressions of political choice.

While Governments operate in the sphere of political choice, trade partners operate in the arena of economic choice. Considering this fundamental difference allows not only to understand systemic incompatibilities, but also to identify common ground. Political choice is much more complex compared to economic choice, as its dimensions cannot be reduced to a single currency.

Characteristics of regulation

Regulation is, by its nature, the attempt to maintain the present or change the future: something is expected to be codified or different after a regulation is in force and applied. In rare cases, regulation may be designed to change the legal assessment of an event in the past (however, international human rights law prohibits such retroactive legislation in most cases). However, regulation will never change the past itself. Both concepts – regulation and risk management – are related to the future. Nevertheless, both concepts build on experience from the past.

For both concepts, it is thus relevant to consider how to adapt to the uncertainty of future events. A rather traditional, common approach is to assess the likelihood of certain events occurring. Another way to manage this challenge is to aim for rather robust, resilient solutions, which are fit for purpose under a variety of possible conditions, rather than accepting the risk of a bet on a certain likelihood materializing. Following these insights, more and more Governments apply strategic foresight for policy programming and regulation. Essentially, strategic foresight accepts that various futures are possible.

Though some political talk may suggest the contrary. Political decisions and resultant regulations are never just right or wrong. Political decisions, resultant regulations and their evaluation always deepen on the preferences of those concerned. Regulation may be based on clear facts and figures and occasionally on assumptions or models that may vary in terms of their accuracy. Regulation may prove to be effective and/or efficient, upon review. A certain regulation might also fit more or less harmoniously in a given legal or real-world context. Nevertheless, when preparing regulation, legislators and policy-makers with different preferences may draw different conclusions from identical facts and figures. This is not a bug of regulatory systems; it is a feature. Particularly in democratic jurisdictions where voters can easily decide for another set of preferences at the next election if they are unhappy with the set of preferences dominating so far. Only international law, e.g., adopted through the United Nations, and usually domestic constitutional frameworks limit the politicians' room for maneuver.

The fact that regulation might be changed over time leads to another attribute: regulation is fluid. Though the way changes are managed might differ from jurisdiction to jurisdiction, all regulation can be changed sooner or later. More easily, its application via implementation and enforcement might also be changed or further developed. For those concerned it is thus essential to stay up to date. On the other hand, the fact that regulation might be changed creates opportunity to influence the evolution of regulations. However, if, how and to whom this option is

available depends heavily on the respective jurisdiction and its type.

Another common myth is the notion that a single jurisdiction's control maybe exclusive. In fact, no policy-maker is an island. The narrative of singular power is illusive in the modern, interconnected and interdependent world. Even if it were possible to identify a location that would be under the exclusive control of a single jurisdiction, the behavior of bordering jurisdictions, mobile forces and global conditions (e.g., on climate), may at least challenge that exclusive control. In consequence, jurisdictions cooperate in modern times in multiple settings internationally, such as via the United Nations Economic Commission for Europe (ECE). Preferably this cooperation is rule-based, since it allows those concerned to make themselves familiar with rules and adapt any requirements.

In addition, most regulatory systems are organized in a multi-level manner. Nearly all jurisdictions practice a decentralized deployment of functions, either for regulation (including implementation and enforcement) or for implementation and/or enforcement alone. Partly, the decentralized deployment of functions enables the legitimate use of discretionary powers by authorities concerned. In this way deployment potentially fosters trade.

Conversely, the deployment of functions may have adverse effects on trade, as it might add complexity to the relevant legal frameworks and might also reduce the clearness and predictability of any given regulation. The significance of these factors depends above all on the coherence of the body of rules across levels. In fact, some jurisdictions are eager to avoid inconsistencies, while other jurisdictions follow a rather competitive approach to regulation. These jurisdictions allow those concerned to choose the regulation with which to comply. Regulatory systems like the latter might appear to be attractive to dominant market actors. However, they add complexity and reduce clarity and predictability for the majority of actors. Their inherent systems risk is therefore higher.

In most legal cultures, regulation follows a two-step-logic: the legal assessment of an event, behavior, product or service depends on both its characteristics and use. For example, a tool like a knife could be perfectly compliant with all legal requirements and standards when considered as a product. However, using or carrying this knife may nevertheless be critical in certain places, at certain times or depending on the purpose of its use.

Limits to both characteristics and use are ideally to be defined in regulations. Thus, ordinarily, an individual product or service is not in itself rendered lawful or unlawful. The lawfulness of a product or service depends on specific conditions and its (intended) use. In consequence, it may be difficult to estimate ex ante whether a product or service complies with legal conditions or not. Confronted with the question of compliance, lawyers and public servants will thus often answer: "it depends." Legally-binding information can in consequence only be given when all conditions are known. The answer "it depends" is in most cases highly unsatisfactory for citizens and businesspeople, as they want to know up front. "It depends" leaves the risk with those concerned… but it grants room for further maneuver.



FIGURE 1: EXAMPLE: POLICY INSTRUMENTS OF THE EUROPEAN COMMISSION

Hard, legally binding rules	• Specific behavior required • Comand-and-control-approach
Soft regulation	• Self-regulation and co-regulation • Technical standards • Pure voluntary bottom-up initiatives (self-regulation) • Legislation-induced co-regulatory
Education and information	• Information and publicity campaigns • Training and guidelines • Disclosure requirements • Standardised testing or rating systems
Economic instruments	• Tradable offsets and permits • Taxes, charges, and fees
Behavioral insights, sandboxes, combinations	• Substitute traditional assumptions by observed evidence • Test innovations in controlled real-world environment, under specific plan, monitored by competent authority • Combinations, e.g. of information disclosure with voluntary initiatives

Source: https://commission.europa.eu/law/law-making-process/planning-and-proposing-law/better-regulation/better-regulation-guidelines-and-toolbox_en; version July 2023, page 122 ff.

Legal practice encompasses numerous ways to structure tools of regulation. Concrete definitions often apply only to specific legal frameworks in defined jurisdictions. Thus, identical expressions might identify very different types of regulation in different jurisdictions. Many concepts refer to rather formal aspects, whether regulation is adopted by the legislature or by the competent executive, whether it creates new substantive rules or provides additional technical detail within the scope of existing laws. Some concepts refer to formal documents (e.g., executive order, decree, ordinance, regulation, directive, etc.), whilst others refer to the rigor of regulation (command and control vs. soft tools; interventionist vs. less intrusive). A comprehensive overview of a wide variety of policy instruments is utilized by the European Commission in its "Better Regulation Toolbox".[1]

Regulatory Reform

Governments aim, by their nature, to be effective. In fact, systems of governance might be considered by those concerned to be more or less effective, efficient, or ambitious. But even if a Government decides to be as small state or non-interventionist as possible, it aims to provide its core functions. These core functions are at the heart of its existence. Vice versa, when Governments lose control, they open the door towards anarchy (the absence of governance) and invite other actors to take over state power. The lack of these core functions, a lack of trust in governance, and a lack of clarity of roles often contribute to fragility of States. However, fragility of States raises risk for trade.

Many Governments have thus decided to commit themselves to regulatory reform. Jurisdictions often undergo a number of different phases when concerned with the quality of their respective regulatory system and its outcomes. Usually, Governments start with structuring and focusing their procedures. The ambition to raise efficiency of regulation at the ground-level has led quite a number of jurisdictions to cutting red tape programs. Often these programs have paved the path for a focus on the quality of evidence, which is used to design and evaluate regulation. More recently, regulatory reform trends have often exploited the potential of collective learning, service design, collaborative formats, qualitative social research and behavioral insights. Examples for the application of all of these methods can be identified in Europe during the past three centuries.

[1] See: https://commission.europa.eu/law/law-making-process/planning-and-proposing-law/better-regulation/better-regulation-guidelines-and-toolbox_en

TABLE 1: ECONOMIC AND POLITICAL CHOICES

	Economic choice	Political choice
Relevant dimension	Monetary value	Various and easily changing political preferences
Number of relevant parties	Distinct players in a specific marketplace	All actors under a specific jurisdiction
Freedom of choice	Individual decision to agree to a transaction or not	Actors under a specific jurisdiction have limited choice
Liability	Individual	General

Source: WP.6

Economic versus political choice

From a business perspective, political systems may appear fuzzy and unpredictable. The reason behind this is that within economic choice environments all aspects can be reduced to a monetary value. Risks linked to goods or services also have a monetary value. Alternatively, this value might be (re-)negotiated between business partners in a specific situation. In any case, the exchange may be agreed upon or not. Within the sphere of economic choice, usually only actors who participate in a particular transaction voluntarily are liable.

Applying rules of mathematics and logic allow similar observations in the sphere of political choice. Monetizing political issues is, however, only possible under a set of specific assumptions and conditions. However, these assumptions and conditions are subject to constant individual change. Calculation models represent reality only for a specific moment. Monetizing political issues is also an artificial exercise, since political issues are not subject to exchange. Finally, the number of players in the political choice sphere is much larger than the number of players interested in acting in a certain marketplace. In addition, all individuals and/or entities under a certain jurisdiction will have to carry the burden of non-beneficial decisions.

While many characteristics of risk management and regulation are similar, the logics of economic and political choice are completely different.

Regulation as Government's answer to perceived risk

The quality of regulation, its implementation and its enforcement depend on many factors beyond this brief overview. In particular, arbitrary decision making, random application of rules, corruption, lack of administrative capacity, unequal treatment of actors depending on size or political preferences and many other malfunctions in public administration may have negative effects on trade and on effective management of risks. Malfunctions of public administration also add to the fragility of jurisdictions.

Nevertheless, both the economic and political management of risks will gain effectiveness, when management of risks aims for robust, clear, predictable and appropriate means, based on evidence and on the ability to learn from experience.

Risk-based market surveillance in practice

It is the task of market surveillance to ensure that products bought and used by consumers are safe and in conformity with essential technical requirements and that all economic operators involved fulfill their legal obligations. In this way we assure confidence in the working of the market and present a level playing field for industry.

However, the sheer number of products that are put on the global market and sold to users and consumers each day is simply enormous. Some are manufactured by local national industry while others are imported from third countries. Therefore, they will have to be inspected in shops, warehouses and at borders, thus covering the complete supply chain.

Although it is solely the obligation and responsibility of the economic operators involved to comply with the law, it is the purpose of market surveillance to encourage this compliance by facilitating good practice and penalizing bad and unsafe shortcuts. For this, the actions of market surveillance authorities (MSAs) need to be visible, relevant and effective – forcing a behavioral change where necessary.

Human and financial resources of MSAs are, by nature, finite. It is therefore necessary, in order to be as effective as possible, to target specific products, sectors and markets, where inspections are either most needed or will have the highest impact.

For products entering the country, a risk-based selection method is applied by customs authorities, taking into account parameters such as the nature of the product, the country of origin, the manufacturer/importer/distributor and their known associates, the shipment method, the size and number of items, and the destination. The exact way in which these parameters are chosen and how they influence the selection process is a closely-guarded secret, known only to a select group within the administration. It is also a dynamic process because of a continuously changing market, but also because the actual selections are closely scrutinized by those parties who want to circumvent them. Continuous adaptation is necessary.



Once a product has entered the country and thus the internal market, it becomes the task of the MSAs to surveil as effective as possible. To achieve this, it is necessary to perform risk based market surveillance based on the broadest possible input from all stakeholders involved and relying on objective, measurable and transparent criteria.

A necessary first step in this process is described in ECE *Recommendation S on Applying Predictive Risk Management Tools for Targeted Market Surveillance.*[2] Recommendation S gives a method for evaluating how dangerous a non-compliant product is, recommending that the evaluation of non-compliance risk should reflect (i) how dangerous a certain product is when it is non-compliant to standards and (ii) what the probability is that a non-compliant product of this type is present on the market.

This approach will result in a theoretical priority list for the surveillance of products, based on their level of danger and probability of being present in the market. However, to establish an inspection plan in practice, MSAs need to take

[2] See: http://www.unece.org/fileadmin/DAM/trade/wp6/Recommendations/Rec_S_en.pdf

several additional considerations into account, which will influence the use of their allocated resources:

- The kind of inspections: pro-active vs re-active.
- The kind of market system involved: open or closed.
- The intended result of the inspections: market study or behavioral change

Furthermore, MSAs work within a specific political, cultural, and societal framework, which often acts to modify the initial risk-based priority list by other external factors, such as:

- Necessity of the product
- Possible affected high-risk groups (elderly, children)
- Threat to free and fair competition
- Size of the market
- Number of known complaints and accidents
- Newly-implemented legislation
- New product or hype
- Results of market studies
- Impact of previous campaigns
- International context
- Media interest
- Political priority

Pro-active vs. re-active inspections

Market surveillance can be "pro-active" or "re-active".

Reactive inspections are carried out in response to an external request or trigger: calibration requests, requests for technical metrological controls, complaints, incidents, accidents, or other. Obviously, it is impossible to estimate in advance exactly what the annual number of reactive inspections will be and in which domain they will take place. For this reason, they are usually not included in an inspection plan. However, a reserve of personnel and resources must be foreseen and provided by the MSA that can be used for this purpose.

Proactive inspections, on the other hand, are always carried out on the MSA's initiative. They can be part of a specific targeted campaign but can also be inspections that are considered necessary and recurrent in general.

Open and closed market systems

When drawing up an inspection plan, it should be taken into account that, depending on the products or services involved, inspections are either carried out in an open market system or in a closed market system.

In a closed market system, all parties and parameters are known (the number and location of companies, the number and nature of products involved, etc.). These are typically heavily-regulated activities (e.g., explosives and fireworks) or a market where there are important post-market requirements (e.g., weighing instruments and fuel pumps). This allows for full statistical monitoring, based on a predetermined frequency of monitoring.

In an open market system, the parties and parameters are not (and never will be) fully known in advance. The inspection plan in this case will be based on priorities to be established and a weighing of available resources against estimated workload.

In both cases a statistical analysis of the sectors or products to be inspected will be made (albeit with diverging levels of certainty) in order to make an estimation of the number of inspections needed to achieve required results with a certain reliability and within a certain margin of error.





Performing market studies or including behavioral changes

It is also necessary to consider whether the purpose of the inspection is to realize a market study or effectively to influence the behavior of the actors within a specific sector.

A market study is performed to verify to what degree non-conformant products are present within a given population and within a predetermined confidence interval. The result can then be used to decide if further steps are necessary: informing the sector, drafting new regulations, or other intervention.

Inspections with the aim to influence behavior seek to create awareness of legal obligations in a specific sector of products, and highlight that inspections are indeed carried out and followed by obligatory measures, penalties and fines. The effectiveness of such campaigns depends upon, among other things, the visibility of the inspections and rigor of any repressive measures.

Input from stakeholders

Any inspection plan needs input from stakeholders. These should be as diverse as possible, both from within the MSA and from actors outside of the MSA.

External stakeholders should be actively contacted and encouraged to introduce proposals for possible products or market segments to be inspected. These stakeholders should represent consumer organizations, sector representatives, media, other ministries and economic operators.

Internal stakeholders should also be considered. Inspectors and other agents of the MSA have first-hand and up-to-date knowledge that represent a very valuable contribution to the inspection plan.

It should be clear that although all proposals will be assessed, these are in effect only proposals, and there is no guarantee that they will be implemented. A valid reason and justification for inclusion will however enhance that possibility. To facilitate the assessment all proposals are normally fitted into a standardized form and summary table.

Further input may be provided by the relevant Minister and his policy team.

Establishing the risk-based inspection plan in practice

The necessary steps to establish a risk-based inspection plan are as follows:

- **Determine the inherent danger** and probability of products, using a predictive risk management tool.
- **Consult stakeholders** to establish a list of products of interest.
- **Evaluate the list of products** of interest, based on societal, cultural and political priorities, using a number of objective factors, including the results of the predictive risk management tool, to create a priority list.
- **Determine for each product** on the priority list the intended kind of market surveillance, taking into account the existing kind of market system, to establish the resources needed.
- **Allocate the available resources** needed for the inspections, according to the priority list.
- **Disregard all subsequent proposals** for which there are no more resources available.

Conclusion

To create an effective and efficient risk-based inspection plan in practice, the MSAs need to reconcile both the priorities identified by the initial risk assessment of the products, and political, cultural and societal priorities, while taking into account their limited resources available.

This can be done by utilizing the results of a predictive risk management tool, as an input to a process that assesses proposals based on objective criteria representing the political, cultural and societal priorities to produce a priority list.

A practical example

Every proposal that results from the stakeholder's input will need to be assessed for each of the chosen criteria, by giving it a score between 0 and 5, with 0 given for being non-relevant for a specific criteria and 5 for being absolutely vital. If all thirteen criteria of the above example are used, this should result in an overall priority score between 0 and 65 (13x5).[3]

All proposals are then ranked according to their overall priority score, giving us a priority list for the inspection plan.

Elements of an inspection

An inspection checks whether a particular product meets the requirements imposed by law. These requirements may be administrative or technical and may relate to quality, safety or metrology. An inspection can sometimes be considered as a check of one specific product or service, or as a check of a number of grouped products or services. This needs to be clearly defined.



An inspection will always consist of one or more of the following actions, depending on the kind of product or installation concerned:[4]

- Travelling to the place of inspection (outward and return)
- Accessing the place of inspection
- Contacting the persons present
- Making findings
- Taking and/or receiving samples
- Performing or attending tests (on site or in a lab)
- Recording the findings on-site (resulting in an inspection report)
- Handing over requested documentation
- Copying documents on site
- Requesting documents from the registry, town hall, police station, or other repository
- Sealing installations on the spot
- Seizing products
- Drafting and issuing warnings on the spot
- Entering data into a database
- Requesting documents necessary for inspection
- Preparing samples for laboratory testing

[3] If not all criteria are deemed to be equally important, the weighting can be diversified by assigning individual multipliers to specific criteria. The higher the multiplier, the more important that criteria. The assessment score for that criteria then needs to be multiplied by the chosen factor before summing up to calculate the overall priority score.

[4] The following are not considered part of an inspection:
i. The preparatory study (determining working methods, drafting procedures, checklists, etc.);
ii. Administrative and judicial follow-up of non-conformities and sanctions imposed.

For each proposal, the necessary number of person-days to perform an inspection needs to be determined, depending on the elements that make up that type of inspection.

Available human resources

Not all employees of an inspection department effectively carry out inspections; managerial and administrative staff do not, in principle, carry out inspections. This should be taken into account when determining the number of full-time equivalents (FTEs) that are effectively available for performing controls.

The workload of the inspectors for performing inspections is on average 55 per cent of their total workload. This is the part of the working time that an inspector carries out an inspection and not preparations, administrative handlings, trainings and meetings.

The number of inspector-FTEs available multiplied by 55 per cent gives the effective number of FTEs available for inspections.

The number of person-days that an FTE represents depends on the country. The multiplication of that number and the effective number of FTEs available for inspections will result in the number of person-days that are available to carry out the inspection plan.

Deciding the type of inspection campaigns

Depending on the type of market (open or closed) and depending on whether the aim is to perform a market study or to influence the market, four types of inspections campaigns are possible:

For each scenario, it should be decided which will be the corresponding type of inspection campaign.

For a closed market system with market study (type I), the parameters for a market study are typically an error rate of 5 per cent, a confidence level of 95 per cent and an expected spread of 50 per cent. The number of inspections to be performed is determined based on the known (or estimated) population.

In an open market system, we have no (or very little) data on the population. For a market study (type II) in this case, we opt to work with a population of 20,000. (In fact statistics demonstrate that there are no significant differences for populations above 20,000.) For a 5 per cent margin of error, a 95 per cent confidence level and an expected spread of 50 per cent, 377 checks need to be performed. Only if there is reliable data to assume that the population is significantly less than 20,000 will the estimated population be used.

Within a closed market with an approach to influence the market (type III), trade-offs will need to be considered, based on the type and nature of the sector and the experience of the inspectors (number of violations already detected, accidents, penalties, etc.), in order to determine the number of actual inspections needed.

Within an open market with an approach to influence the market (type IV), trade-offs will need to be considered based on the type and nature of the sector and the experience of the inspectors (number of violations already detected, accidents, penalties, etc.) in order to determine the number of actual inspections needed.

Finalizing the inspection plan

In order of priority, depending on the type of inspection campaign, the number of inspections to be performed for each proposal is determined. The number of inspections for a proposal is then multiplied by number of person-days required to perform an inspection. The result is the total number of person-days needed for that campaign. Beginning with the proposal that has received the highest priority score, the necessary total number of person-days is assigned and deducted from the total number of departmental person-days available, until the available person-days are exhausted. Together these proposals make up the annual inspection plan. No further proposals should normally be considered.

TABLE 2: CLOSED VS. OPEN MARKET SYSTEM

	Closed market system	**Open** market system
Market Study	**type I**	**type II**
Influencing the market	**type III**	**type IV**

Source: WP.6

Risk management **in border clearance**



The volume of trade, the speedy and shortened time of delivery of cargo, the need to facilitate trade to enable economic growth and the reduced capacity to employ a large number of human resources and strains in financial budgets are challenges that currently confront border regulatory agencies involved in cross border cargo clearance. Hence, the need to adopt risk management has never been more important, as enhancement in technology is transforming the way trade is conducted and continually improving delivery of cargo.

There is often more than one regulatory agency at the border in many countries, all with different objectives, which means that the risks being considered vary from agency to agency. For instance, while customs' objectives revolve around collection of duties through proper valuation, classification and determination of origin of goods, another agency like the food and drugs or the standards organizations would be concerned mostly about the safety of food and drugs, and quality assurance of products coming into a country.

The use of risk management therefore has to be adopted in such a way that the risks of all border agencies are addressed in a harmonized manner to ensure trade is facilitated, for instance, in a single window system. A harmonized or centralized risk management approach will provide assurance to all border agencies and engender cooperation and data sharing amongst them.

Nevertheless, for the purpose of this article, the focus will be limited to the use of risk management in border clearance at the customs level.

Customs role in market access

In many countries, customs often takes the lead in cargo clearance and is therefore saddled with the duty of facilitating border clearance. In fact, in many national trade facilitation bodies around the world, customs is often identified as the agency responsible for facilitating trade at the borders. Naturally, customs, in a bid to carry out its traditional role of duties collection and combating smuggling of illicit goods across borders, often tends to focus on the need to control the flow of goods, while neglecting or paying less attention to the need to facilitate movement of legitimate goods across the border.

As a result of this, the World Customs Organization (WCO) has over the years developed documents and tools which recommend the use of risk management to balance control and facilitation. Examples of these documents include the *Risk Management Compendium (Vol 1 & 2)*[5] and the *Revised Kyoto Convention*[6]. Several other world bodies like UN/CEFACT, the Organisation for Economic Co-operation and Development (OECD) and World Bank have also developed documents to help customs administrations adopt and implement risk management in the most effective way.

The core functions of customs are similar in many countries around the world, though the aspect of focus may differ. These functions include collection of duties and taxes, providing security at the borders, prevention of smuggling of illicit goods into the country, and collection of data. In some developed Nations for instance, a lot of focus has shifted from collection of duties to the aspect of security. An example of this can be seen in the signing of the WCO *Punta Cana Resolution*[7] in 2015, which urged customs administrations around the world to give greater attention to their role in national security as a result of the rise in terrorism. In developing countries, greater emphasis is laid on the collections of duties and taxes, which is oftentimes, a huge source of revenue to fund national budgets.

[5] See: https://www.wcoomd.org/en/topics/enforcement-and-compliance/instruments-and-tools/compendiums/rmc.aspx

[6] See: https://www.wcoomd.org/en/topics/facilitation/instrument-and-tools/conventions/pf_revised_kyoto_conv.aspx

[7] See: https://www.wcoomd.org/en/media/newsroom/2015/december/wco-issues-the-punta-cana-resolution.aspx

Risk management in customs

Risk indicators can include aspects pertaining to the following. These are often used as pointers for risk identification. A combination of several indicators is required during analysis to be able to draw a reasonable conclusion. The data generated relating to these indicators could be used in automated systems to ensure profiles are created and monitored.

- The importer or consignee
- The exporter or consignor
- The shipping agent
- The cargo handler
- The licensed customs broker
- The product
- The container
- The transportation
- The carrier's manifest
- The bill of lading
- The air waybill
- The countries of routing
- The e-commerce platform

Risk profiling

After identifying the areas of risks, it is important to apply risk profiles. This is an anticipatory procedure that is used to record all risks perceived in a specific field and classifies them into high, medium and low.

"A profile is a set of characteristics, some or all of which tend to be present whatever the specific circumstances may be, and which are common to the shipment of goods and travelers involved in smuggling activities or customs fraud at a given location."[8]

To achieve maximum result using a risk profile, there are six (6) phases required, which are

1. Collection of available data
2. Structured representation of information
3. Data analysis
4. Distribution of profile
5. Feedback
6. Update and revision

By using profiles developed on the basis of risk indicators and relevant criteria, it is possible to reconcile the demands of control, monitoring and facilitating trade effectively.

Risk selection

This is the process whereby risk indicators that have been used in profiling, are used to target goods, passengers, and means of transport, for the purpose of their inspection by the customs authorities.

This selectivity assists with facilitating legitimate traffic of cross border cargo and monitoring suspected traffic of illicit goods. Hence the dilemma faced by customs: whether to focus on facilitation or control.

Conclusion

By using profiles developed on the basis of risk indicators and relevant criteria, it is possible to reconcile the demands of monitoring and facilitating trade effectively.

It is important to note that applying an intelligence-driven risk management which relies solely on the use of data will enhance the result of targeting and increase efficiency of limited resources.

Customs administrations may also need to be aware and consider the use of risk management in areas that could indirectly influence the attainment of their core objectives, especially as it relates to other regulatory partner agencies and private stakeholders to enable economic growth.



[8] World Customs Organization Risk Profile Definition accordeon to the WCO *Risk Management Compendium*.

Risk management **in conformity assessment**

In many regulatory systems, safety is defined in terms of a tolerable level of risk, and the level of risk of a product which is compliant with regulatory requirements is deemed to be tolerable. Conformity assessment involves activities aimed at demonstrating compliance with specified requirements; thus, it plays a key role in ensuring that products placed on the market are safe.



Conformity assessment may apply to products, processes, services, systems, installations, data, designs, materials, persons, bodies or organizations (or any combination thereof). It provides assurance that the requirements set out in specifications such as international, regional or national standards, as well as guidelines, recommendations or other normative documents are met.

According to international best practice – including the World Trade Organization's (WTO) *Technical Barriers to Trade (TBT) Agreement*[9] and the ECE *Recommendation L on an International Model for Transnational Regulatory Cooperation Based on Good Regulatory Practice*[10] – technical regulations are based on international standards. Compliance with relevant standards is hence a way to demonstrate compliance with regulations. Within the regulatory context, conformity assessment refers to pre-market controls, which, together with the relevant processes, is one of the main building blocks of any regulatory system.

The objective of conformity assessment in a regulatory system is to prevent non-conforming products – products with a level of risk might which is not tolerable – from being placed on the market, while avoiding unnecessary costs on the part of economic operators and promoting the removal of technical barriers to trade. This is the principle "tested once – accepted everywhere."

Pre-market control in various forms is implemented in practically all markets. In some spheres, such as finance, conformity assessment is conducted through licensing. In technical regulation, it is performed as part of a set of complex processes involving international players and based on internationally-recognized standards.

Demonstrating compliance with standards and other criteria is becoming increasingly important not only from the regulatory perspective but also to consumer confidence, as products and services become more technically sophisticated, and their use becomes more widespread around the world. It is becoming more difficult to evaluate the probability and consequences of the non-compliance of products and services.

Thus, conformity assessment is an integral part of global trade, business, standardization, recognition and compliance.

The relevance of climate change is now necessarily considered in management systems and accordingly included in mitigation goals and actions; this leads to the identification of relevant risk factors.

Choosing conformity assessment methods

The main conformity assessment methods and their most common applications include testing, inspection, assessment, examination, audit, assessment, declaration, certification, accreditation, peer review, verification and validation. The conformity assessment methods are often prescribed and belong to a particular mandatory (within

[9] See: https://www.wto.org/english/tratop_e/tbt_e/tbt_e.htm

[10] See: https://unece.org/DAM/trade/wp6/Recommendations/Recommendation_L_en.pdf



a regulatory context) or voluntary conformity assessment scheme.

Confirmation of conformity as the issuance of a statement based on a decision can be carried out by a first party (self-assessment by the manufacturer or supplier), a second party (buyer or user) or a third party (testing laboratory, certification body, control body).

From the risk management perspective, it is crucial to ensure that conformity assessment schemes for products are proportionate to the consequences and probability of non-compliance of these products. This principle is emphasized in various recommendations and international agreements; for example, the WTO *TBT Agreement* emphasizes that "conformity assessment procedures shall not be more strict or be applied more strictly than is necessary to give the importing member adequate confidence that products conform with the applicable technical regulations or standards, taking account of the risks non-conformity would create."[11] This, in turn, requires using the concept of non-compliance risk of a product when choosing conformity assessment schemes.

The time and money spent on compliance assessment must be balanced against the risks of non-conformity. Market acceptance of adequate costs and acceptable levels of risk must also be taken into account. That is why planning the scope, volume, complexity, timing and, accordingly, the cost of conformity assessment should be based on an assessment of the risks that may arise as a result of the non-compliance of the assessed object with the established requirement, taking into account the likelihood and consequences of such non-compliance.

Risk management and conformity marking: the CE marking example

Within the European Union (EU), many products require a "CE" marking before they can be sold. The CE marking indicates that a product has been assessed by the manufacturer and found to comply with EU safety, health and environmental regulations. This is required for products manufactured anywhere in the world that are then sold in the EU.

To attach the CE mark to its product, a manufacturer must complete a number of steps, including selecting directives, checking product requirements, determining the need for an independent assessment, testing the products, preparing the necessary technical documentation and labeling, and issuing a declaration. These steps may vary depending on the product as the conformity assessment procedure varies. This is only mandatory for some products covered by the EU New Approach Directives.[12]

Risk classification here is achieved by defining quantitative and qualitative risk assessment criteria. Once risks have been identified and designated as risk types, an assessment of the inherent and residual risks is carried out, taking into account the level of controls in place to mitigate the risks. For example, medical devices are divided into various risk classes: classes I to III for general medical devices and classes A to D for in vitro diagnostic medical devices. Class I and Class A present the lowest risk, while Class III and Class D present the highest risk.

The main risks of conformity assessment: safety risk vs. economic risk

Many regulatory frameworks contain requirements for economic operators to conduct a proper risk analysis of products and services. Such risk analysis could be a part of conformity assessment. Also, conformity assessment itself may lead to incorrect decisions, the risks of which should be anticipated and taken into account. This risk may be single or specific, where a particular product is incorrectly accepted or rejected. It can also be a global or general risk: the part or proportion of goods that may be incorrectly accepted or rejected. These risks can be borne by the consumer, when a nonconforming product is mistakenly accepted (a safety risk), or by the manufacturer, when a conforming product is mistakenly rejected (an economic risk).

[11] Op.cit. article 5.1.2

[12] See: https://eur-lex.europa.eu/EN/legal-content/summary/enhancing-the-implementation-of-the-new-approach-directives.html

Integrated risk management **in border control**



In the realm of border control and import compliance, risk management is only as strong as its weakest link. In practice, borders are very busy areas; in some countries, as many as 25 regulatory agencies can be involved in border control, sometimes inspecting a single shipment.[13] And while customs authorities typically take center stage in border management, they are just one component of a larger network of agencies involved in the process. A deficiency in any border regulatory agency's capacity can jeopardize the entire system's efficiency.

A comprehensive risk management approach and strategy is therefore imperative to enhance border control efficiency and facilitate international trade. Harmonizing risk evaluation criteria across regulatory agencies is equally essential; without such alignment, efforts to manage risks may prove ineffective in reducing time and costs related to border compliance.

Challenges in effective risk management

Despite its significance, implementing effective risk management within trade procedures remains a considerable challenge globally.[14] In fact, the WTO *Trade Facilitation Agreement (TFA)* Database showed in 2023 that risk management was one of the five measures with the lowest implementation rate globally. Many countries struggle with fully realizing the potential benefits of risk management systems, often experiencing stagnant border compliance procedures years after system implementation.

Challenges predominantly revolve around building the risk management capacity of regulatory agencies and integrating risk management procedures across all agencies involved in border control. In the case of lack of capacity, if evaluations of incoming shipments are biased or incomplete, or if no risk criteria based on regulatory objectives are established, risk management will result in mitigation measures that are not proportionate to the risks they were set out to address. The second set of challenges is related to the disparate nature of risk management procedures, with each agency responsible for its own set of non-compliance risks associated with every incoming shipment.

Objectives and benefits of building a national strategy

Improving border control efficiency requires the adoption of a comprehensive risk management strategy, including an integrated risk-based import compliance framework. Such a strategy, which aims to facilitate trade while protecting the health and safety of consumers, society and the environment, aligns with international trade principles outlined in WTO Agreements – in particular the *TFA*, the *TBT Agreement* and *Sanitary and Phytosanitary Measures (SPS) Agreement* – and prioritizes non-compliance risks based on severity and probability.

[13] See: OSCE/UNECE (2012). *Handbook of Best Practices at Border Crossings – A Trade and Transport Facilitation Perspective.*

[14] See: WTO. *Trade Facilitation Agreement Database*. Data retrieved as of March 2023 from https://tfadatabase.org/implementation/progress-by-measure

An integrated approach, which usually implies integrating the risk management processes of regulatory agencies in the system of customs, results in a comprehensive overview of risks. Strengthening import compliance enhances market surveillance and enforcement activities, fostering collaborative border management. Finally, an integrated approach not only streamlines risk management processes but also optimizes resource utilization by enabling knowledge and infrastructure sharing among regulatory agencies.

Implementation roadmap

Detailed guidance on developing a strategy and implementation plan for integrated risk management as well as international practices have been outlined in a dedicated publication by the International Trade Centre (ITC) and the ECE, *Managing Risk for Safe, Efficient Trade: Guide for Border Regulators.*[15] This chapter provides a summary of its key recommendations.

Integrating the risk management systems of various regulatory authorities in a single framework covers all processes and elements of a risk targeting system, from developing compliance rules to preforming inspections. An integrated risk management system aims to optimize resources, remove duplicate functions and apply standardized methodologies.

To this end, there are tools that all regulatory authorities involved in border control could use to develop compliance rules and build risk profiles to assess every incoming shipment, as well as to evaluate them based on risk tolerance. There are also specific steps that a regulatory authority must take to apply these compliance rules and to choose a sampling plan that is proportionate to the level of the non-compliance risk of a shipment.

The roadmap for building integrated import compliance systems encompasses the following layers:

- **Organizational level:** implementation of formal risk management within regulatory agencies.
- **Regulatory system level:** ensuring that relevant regulatory systems – those containing border control as a building block – are risk-based and support the Sustainable Development Goals (SDGs), particularly in advancing safety of international trade.
- **Import compliance level** of a regulatory agency: applying profiling and targeting techniques in regulatory agencies responsible for border control to evaluate the non-compliance risk of incoming shipments.
- **Integration level:** integrating the import compliance systems of regulatory agencies involved in border control.

Establishing formal risk management practices within regulatory agencies forms the foundation of this roadmap, aiming to promote a shared understanding of risk management methodologies among all stakeholders involved in international trade.

The key challenge to integration is linked to the complexity of projects and number of stakeholders needed to cooperate and implement it. Customs authorities are often the leading agencies in integration projects and, in most cases, their risk management system is used as a basis for integration.

To cooperate efficiently with border agencies, customs authorities are advised to do the following: (i) develop agreements and mechanisms for intra-organizational risk-assessment, intelligence sharing and conduct of coordinated and joint cross-border controls, (ii) share the infrastructure for inspections, and (iii) create information-exchange mechanisms and joint operating procedures.

Business Engagement

The involvement of businesses is crucial in advancing integrated risk management efforts. As clients of integrated systems, businesses should partner by cooperating and engaging with border regulatory authorities in an ongoing dialogue and advocacy. They should further promote compliance by investing in internal reforms and applying best practices.

Conclusion

Effective risk management in border control and import compliance is paramount for facilitating international trade while safeguarding regulatory objectives. By addressing challenges and adopting an integrated approach, countries can enhance border control efficiency, reduce compliance burdens, and foster smoother and safer international trade operations.

[15] International Trade Centre (2022). *Managing Risk for Safe, Efficient Trade.* ITC, Geneva.

Sector risk and business risk
in risk-based inspection systems

The risk profile of a business

The function of implementing laws and regulations should be risk-based. When inspectors go out to inspect compliance with regulations, they will do so according to a plan which has mapped the risk landscape of their sector, not according to the physical proximity with the inspector. Inspection resources will be deployed where they will be most effective; this will be where the risks of dangerous non-compliance are greatest. The risk landscape will be drawn according to two types of risk – the risks coming from the sector (sector risk) and the business itself (business risk).

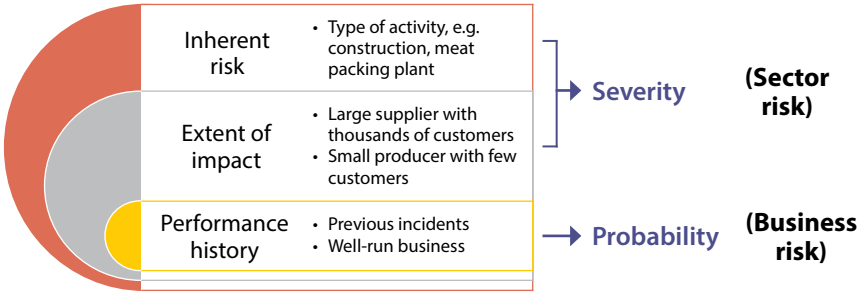


Sector risk covers the real-world dangers that need to be managed in that sector of the economy, such as food-borne illness, workplace accidents, unsafe products, unfair competition, or damage to the environment. In each sector, the responsible ministry and its delivery agency will have identified and prioritized the key risks and there will usually be laws and regulations that set out what activities are to be discouraged or encouraged in order to manage these risks. These rules will usually be aimed at what businesses do.

Business risk covers the performance of the businesses that operate in that sector. These are the subjects of inspection. It is the businesses that control the relevant sector risks – or fail to control them. A nuclear power station will always be a high-risk business, because of the immense sector risk, but it should also be well run so that these risks are properly managed. On the other hand, careless or incompetent businesses can add risk through their actions.

These two risks become combined in a risk profile for each business. The two types of risk complement each other insofar as the sector risk determines

FIGURE 2: RISK PROFILE OF A BUSINESS



Source: WP.6



the severity of the risk and the business risk determines the probability that such a risk will happen, as in figure 2.

By building a risk profile for a business, it can be given a classification – high, medium or low risk – or even a score and ranking that places it relative to all other businesses in the sector. In Albania, their central inspectorate has data on all businesses derived from their inspection record over many years. Taking the number of violations found in an inspection of that business over the years as the metric, it can rank all businesses in terms of its performance history, from those with the fewest violations to those with the most. This metric will be combined with that business's score on sector risk to give its overall risk profile score. It may have an enormous number of violations but if it is a small business whose dangerous management affects very few people, it may still be a low-risk business overall and not worth many inspections. If it has a high sector risk score but an exemplary compliance history, it may not be classified as high risk.

Developing the methodology

There is much more to market surveillance than inspecting products. The risk that comes from businesses is also very important and can be underrated. The product safety system relies on pre-market processes and systems that ensure that products will be safe for intended use when compliant and that safety level should, in theory still apply for the life cycle of the product. Arguably, the biggest danger to this system comes from the actions of businesses that are involved with the products all along the supply chain, from production to retail. In contrast, the safety level of food products can be determined by external factors such as pathogens, temperature or cross-contamination. Manufactured products tend to be more reliable and stable. It is usually human intervention that introduces non-compliance. When applying risk to market surveillance the probability of non-compliance is primarily a matter of assessing the business and how well it manages the probability of non-compliant products. As a simple example, does a retail business check that all its products have a conformity assessment mark, such as the EU's CE mark, or does it not care?

In 2004 in England, the Local Authority Coordinators of Regulatory Services (LACORS) published a risk assessment methodology[16] for Trading Standards Officers (market surveillance inspectors) to apply when planning inspections for market surveillance. They had realized that the supply chain for products could affect the management of product risk. For example, a business that imported in bulk, but then divided and re-packed products, was a separate source of risk. It developed a methodology that involved:

- **A typology** of businesses according to their sector and activities, e.g., B2 – local retailer of regulated goods with weighing equipment, or B9 – Solid Fuel/Bulk Fuel Retailer.
- **The national element** – this covered the potential risk of types of businesses as a category, with a scoring system according to various factors.
- **The local element** – this covered the potential risk of specific businesses and this assessment would come from the experience and observation of inspectors; this too had a scoring system.
- **An overall score** – this could then be used to plan inspections on the basis of the risk presented by each business.

[16] See, for example the Trading Standards Risk Assessment Scheme: https://www.caerphilly.gov.uk/caerphillydocs/trading-standards/trading_standards_ra_scheme.aspx

This was superseded in 2012 by a methodology developed by the Better Regulation Delivery Office (BRDO).[17] It came out of a study of risk assessments methods that were being applied in different regulatory regimes which looked at whether they could be combined into a common risk assessment methodology for inspections and enforcement. Their review concluded fairly easily that methods for assessing sector risk needed to be specific to these sectors, and there was little value in trying to develop a common approach. On the other hand, for business risk, they came to the opposite conclusion. The characteristics of a business will be much the same, regardless of what regulatory regimes govern it for different activities. If it is a chaotic, cash-based family business, it will operate that business model whether it is selling second-hand clothes or frozen food. If it is a systems-based multinational with a legal division, a compliance division and a risk division, it will apply the same resources and skill to all its activities.

So, BRDO produced its own methodology for business risk but referred to it as likelihood of compliance (LoC). It is basically the same as figure 2, where the business risk is the probability element, which is the likelihood of compliance. For assessing the hazard, BRDO admits that this is a matter of sector risk and needs to be determined by whatever methods are used in each sector. They set out four common criteria for assessing LoC: track record/history of compliance, current level of compliance, management assessment (manager's competence, technical knowledge, leadership and commitment to compliance), and management systems assessment (presence and implementation of effective systems for managing risks). They then provide guidance on how to convert various factors into levels of probability of compliance. This builds on the four common criteria mentioned above.[18]

- **Very high:** Minimal number of justified complaints/preventable incidents (taking into account the size of the business); good track record of compliance and responds positively to address any problems identified by enforcement officers, or other sources; current level of compliance is very good with any non-compliance being of only minor significance; knowledge of hazards and how to control risks are embedded throughout the organization; management are fully competent to manage the risks with effective training and development arrangements in place to ensure this remains the case; the business is aware of where external technical knowledge may be required and uses it (where necessary); management are demonstrating strong leadership towards and commitment to regulatory compliance; systems used for managing risks are effective, implemented, monitored, documented, reviewed, and subject to external audit/certification by an accredited organization; the business is demonstrating that it can effectively self-regulate.
- **High:** Few justified complaints/preventable incidents; good track record of compliance and generally responds positively to address problems identified by enforcement officers or other sources; current level of compliance is good; good appreciation of hazards and the associated legal requirements; management have the necessary competence to manage the most significant risks; external technical knowledge is used (where necessary); management are demonstrating leadership towards and commitment to regulatory compliance; systems used for managing risks are satisfactory, e.g., they are fully documented but may not be subject to external audit/certification by an accredited organization.
- **Medium:** Some (though not excessive) justified complaints/preventable incidents; reasonable track record of compliance and response to addressing problems identified by enforcement officers or other sources tends to be positive; currently some non-compliance but not of significant concern; some appreciation of hazards and knowledge of legal requirements; management demonstrate reasonable level of competence in managing risks; external technical knowledge is occasionally used (where needed); some evidence of leadership towards and commitment to regulatory compliance; systems exist for managing risks but require improving in some respects, e.g., may not be fully documented and are unlikely to be subject to external audit/certification.
- **Low:** Significant numbers of justified complaints/preventable incidents; patchy track record of compliance; currently some significant non-compliance; response to addressing problems identified by enforcement officers or other sources is variable and can be poor; appreciation of hazards and associated legal requirements is not adequate; management competence lacking in a number of respects; limited use of external technical knowledge; little evidence of leadership towards and commitment to regulatory compliance; systems used for managing risks are largely ineffective, e.g., unlikely to be fully documented and will not be subject to external audit/accreditation.
- **Very low:** Large numbers of justified complaints/preventable incidents; very poor track record of compliance; current level of compliance very poor; response to addressing problems identified by enforcement officers or other sources is generally lacking and ineffectual; very poor appreciation of hazards and associated legal requirements; management lack

[17] See: https://assets.publishing.service.gov.uk/media/5a7c976a40f0b6629523a82e/risk-assessment-paper.pdf

[18] See: https://assets.publishing.service.gov.uk/media/5a7c976a40f0b6629523a82e/risk-assessment-paper.pdf

the basic competences needed to manage risks; no use made of external technical knowledge; no evidence of leadership towards or commitment to regulatory compliance; systems used for managing risks either non-existent or wholly inadequate.

At the same time as BRDO published this approach to business risk, another risk methodology was published by the European Union Network for the Implementation and Enforcement of Environmental Law (IMPEL), called the integrated risk assessment methodology (IRAM).[19] It is based on a combination of sector risk and business risk, which it refers to as impact criteria and operator performance criteria. The methodology requires weighting factors for both sets of criteria and proposes a few in its guidance book.

This approach requires some subjective assessment by the inspectors, especially in relation to the competence of the business' management. In the system of the United Kingdom of Great Britain and Northern Ireland, there is enough confidence in the professionalism of the inspectors to operate this system. Other countries have greater concerns in balancing effective inspection with anti-corruption concerns about allowing subjective judgements by inspectors. This has led to other approaches to assessing business risk.

Armenia has been operating a risk-based inspection (RBI) system since 2013 across all its six inspection bodies (IBs). It is a very mature system with its own unique characteristics in the way it has developed the original approach of RBI. In the early days, it had concerns about combating corruption or even the possibilities of corruption, and so tried to reduce subjective assessment by inspectors as much as it could. It has also been trying to develop common risk assessment systems across its various IBs, through the work of its Inspection Bodies' Coordination Bureau (IBCB), and came to the same conclusion as BRDO, i.e., that sector risk has to remain within the sectors, but business risk can have a common methodology. What it has developed is a methodology with many variables that are under the control of each IB but operate under a common methodology. For them, business risk has five factors, each with a weighted score out of 100:

1. The **first factor** is about the activity carried out by the business and the type of processes or systems it applies in doing so. The information for this may come from external sources if there has not been an inspection at this point but otherwise comes from the inspection. It is up to each IB to decide what information it gathers to build this account of the business.
1. The **second factor** is based on the results of the checklists where there have been inspections and records the compliance levels demonstrated by the business.
1. If the business has been given an improvement notice after an inspection, this **third factor** assesses how fully and promptly it responded to the notice, or whether it delayed or avoided taking action.
1. If there has been an unplanned inspection as a result of a complaint or a request from another Government body, it may not have applied the full checklist of a normal inspection, so a separate **fourth factor** is assigned to assessing the result of that inspection.
1. The **final factor** is the frequency and repetition of violations.



[19] https://www.impel.eu/en/tools/easy-tools-risk-assessment-guidance-book

These factors are objectively verifiable and present a more detailed picture of the business's performance than the UK methodologies. The weighting or scoring system is quite complicated since a varying number of these factors may be relevant to the case, such as the case where a new business has not had any inspections, therefore only the first applies. There are also rules on how to deal with corporate structures, such as franchises, chains or branches.

The emphasis in the Armenian system on the diversity of approaches by different IBs goes against what is seen as a strength of greater uniformity in other countries developing business risk systems, namely that they can combine research and data across different regulatory agencies. If they are inspecting the same businesses, they can share their business risk data and build an RBI system faster than each one developing from scratch. But the maturity of the Armenian system perhaps doesn't need that advantage.

Part 2:

Risk management **in businesses**

Integrating regulatory risk into a **business' risk management system**



Having discussed risk management in regulatory systems, it is important to consider how businesses view regulatory risk. Whilst regulation, inspections and sanctions may influence the behavior of private sector entities, businesses themselves often approach these regulatory measures as one part of a broader assessment of risks to their business, often by applying their own, internal risk management processes. This section examines how businesses approach risk management, including how the measures discussed in previous sections are often perceived by private sector entities.

Understanding a business' approach to risk and compliance

No matter the industry, location, or business size, there are a myriad of challenges and business risks facing companies in today's dynamic business landscape. Simply put, business risk encompasses any event or situation with the capacity to hinder a company from attaining its business goals, objectives or to continue operating. These risks may originate internally, such as within the company's strategy or industry, or externally, influenced by factors like the global economy, third parties or new government legislation.

This section reviews the different types of risks businesses face and how businesses can manage these effectively.

Categories of business risk

First, it is essential to understand how each type of risk develops and what it means in practice for the business – after all, each business or organization will have a unique risk landscape to contend with. Then, when the business' risks are assessed and categorized, the next step is to address each with a comprehensive risk management strategy. There are several types of risk, including but not limited to the examples below:

- **Strategic risk:** Strategic risk pertains to specific internal or external occurrences that could hinder a company's ability to achieve its objectives and strategic goals. Consequently, such events may necessitate a company's strategic direction shift.
- **Compliance risk:** When new rules or government legislation that may impact the business are introduced, this results in compliance-related or regulatory risk. Depending on the country, industry or employee count, businesses face various regulations to comply with. Some recently introduced regulations within the EU for example include the *EU Whistleblower Protection Directive*, *EU Artificial Intelligence Act or EU Corporate Sustainability Due Diligence Directive (CSDDD)*.
- **Financial risk:** Financial risks are related to the ability of the business to maintain a sufficient cash flow; this includes risks like increased interest rates on business loans, unexpected large expenses, or dealing with non-paying customers.
- **Operational risk:** Operational risk often arises from unsuccessful processes, human errors, system malfunctions, or external occurrences such as natural disasters. This category covers incidents like employee mistakes, technological glitches, and natural and human-induced disasters.
- **Cybersecurity risk:** Cybersecurity threats are a type of operational risk which primarily target computer systems and data. These threats can come in the forms of malware, phishing and malicious scams. These malicious scams typically aim to

access, change, or destroy sensitive information, extort money from users via ransomware, or interrupt normal business processes.

- **Third-party risk:** Third-party risk involves issues associated with utilizing external entities, often referred to as vendors, suppliers, partners, contractors, or service providers. This risk can cause unexpected outages in the supply chain or other compromised business processes, both often resulting in financial loss or business continuity issues (linking back to financial risk).
- **Reputational risk:** Reputational risk threatens a company's positive standing and image and can happen due to different types of incidents, such as rightly or wrongly exposing wrongdoing or unethical practices, greenwashing, or enforcement action. Such risks are complicated for businesses as they often erode stakeholders' trust in the organization. When faced with a reputational incident, customers might express adverse public sentiments about the company, which can lead to a substantial profit loss.

Apart from the main risk categories listed above, various other factors pose threats to a business, such as:

- **Environmental risk:** This includes challenges like natural disasters and events related to climate change. Increased demand from regulators, investors, the public and employees are also driving the need to address environmental sustainability. Environmental, social and governance (ESG) is a hotly-debated issue for many companies, countries and persons, but the incoming wave of regulation such as the aforementioned CSDDD and the *Corporate Sustainability Reporting Directive (CSRD)* coming out of the EU will have global reach and result in both ESG-related risk and compliance risk.
- **Political and economic instability:** This is particularly relevant when exporting goods to foreign markets. It can be linked to working with partners in unstable areas or facing sanctions.
- **Health and safety risk:** Health and safety risks are when employees or people within the supply chains are exposed to potential hazards such as dangerous equipment, harmful substances, fires or mental stress. It can also include a health pandemic crisis.
- **Commercial risk:** This involves risks like the failure of customers to pick up the products or services proposed by the business.
- **Workforce risk:** This includes challenges like maintaining an adequate number of staff, ensuring employee safety, and keeping skills up to date.

It is important to note these risk categories are not fixed, and business risks may fall into multiple categories. For instance, risks related to data protection might be considered when evaluating operational processes, cybersecurity, and compliance.



The use of integrated technology

Given the many risks and compliance issues facing businesses, addressing them individually is unfeasible. The ever-changing nature of business risks challenges companies to keep up with ongoing developments. Consequently, many enterprises utilize advanced technology to implement a comprehensive governance, risk and compliance initiative, rather than treating each risk and compliance issue in isolation.

In order to keep up with the growing complexity of risk and compliance requirements, companies often utilize integrated regulatory management systems that proactively anticipates, monitors and facilitates the management of all risk areas within the business. This avoids the practice of isolating business risks in silos, where different departments manage different areas of risk on separate systems. Critical sectors such as finance, healthcare and transport often have the strictest legislations and risks to navigate, often making them the first companies to adapt to the newest processes of risk management.

In addition to using an integrated system, companies can lean on external parties. The UNECE Working Party on Regulatory Cooperation and Standardization Policies (WP.6) is a forum for dialogue among regulators and policy makers. Participants discuss, make recommendations and develop best practices on a wide range of issues, including regulatory cooperation. The Group of Experts on Risk Management in Regulatory Systems (GRM) is a focal point within WP.6 on this topic. The ECE *Recommendation V on Addressing Product Non-Compliance Risk in International Trade* provides a model for integrated risk management which can be applied to businesses. It has also developed ECE *Recommendation P on Crisis Management in Regulatory Systems* which can be applied to the health risks businesses and administrations face during a major incident such as the COVID-19 crisis. This guidance and the exchange of best practices can make it easier for companies to navigate through the patchwork of regulations.

The benefits of businesses using advanced compliance system

A comprehensive regulatory compliance management system can assist companies to deal with compliance and other risks. Such a system should regularly monitor global and industry regulations to ensure that necessary changes are made to the system and companies' policies and procedures. Doing so can help avoid potential risks and build a document repository for future reference. It can also help to define the companies' long-term goals and consider yearly or biannual audits of the organizations' regulatory compliance strategy and system to gauge its effectiveness and make needed adjustments.

The ongoing monitoring and tracking of employee violations can help to respond to potential reputational risks and workforce risks. Employees should be held accountable. In order to respond to operational risks, the system can track whether employees have seen and attested to the companies' policies and procedures to try to avoid violations.

Ideally, businesses' risk management systems should be automated. This can help streamline the manual work required to comply with regulations, workflows, reports and documentation. By doing so, an organization can save a huge amount of employee time. Multiple automated systems exist to cater to different types of companies.

Such a system may provide a broader field of vision to open up new business opportunities. Organizations can proactively build the necessary resources to navigate challenges and find practical solutions by looking beyond immediate achievements and anticipating potential obstacles. In addition, a third-party platform can help to add an external gap analysis assessment and discover the pain points in achieving risk management per company. A robust risk management strategy and system minimize disruptions, ensuring smooth and efficient operations, allowing an organization to operate more efficiently and provide operational resilience.

Stakeholder trustworthiness is often built gradually for businesses. Acknowledging and planning for risks most likely to impact a business demonstrates resilience and long-term viability, earning the trust of stakeholders. Proactively discussing these risks fosters trust more effectively than reacting to an unexpected crisis.

Whether taking small steps or significant strides towards a greener future, the tightening of regulations and the global shift towards environmental consciousness is undeniable. Companies can take a proactive sustainability stance. Identifying and addressing environmental and regulatory risks positions an organization ahead, reducing its corporate footprint.

In today's business landscape, companies that take no risks will fail to exist. However, taking calculated risks requires an understanding of a business' overall risk profile. Therefore, organizations that use advanced regulatory systems will safeguard their operations, harmonize with the regulatory environment and position themselves for sustainable growth. In addition, advanced technology will establish legal regulations as the foundation of doing business, rather than the ceiling, and provide employees with more time to focus on strategic work rather than ticking boxes to ensure a company remains fine free. The era of siloed risk management being an acceptable solution is fading, making way for a more holistic, integrated, and forward-looking approach that aligns with the demands of digital transformation and ensures a safe and profitable journey forward.

Navigating risks in the international supply chain: **safeguarding global trade**



The international supply chain and global value chains in general represent intricate interconnections of stakeholders that can span across multiple continents and countries. This process includes bringing raw materials to manufacturers (there could be multiple manufacturers on a single value chain) who develop products which are then transported to distribution sites and then brought to consumers. Throughout this chain, there is a constant necessity of moving materials or goods, making global transport and logistics a core element.

Global transport and logistics networks contribute to improving market access, increasing supply chain efficiency, facilitating trade, boosting economic development, promoting regional integration, supporting environmental sustainability, and directly contributing to the United Nations Sustainable Development Goals (SDGs). In particular, they contribute to SDG 8 on decent work and economic growth, SDG 9 on industry, innovation and infrastructure, SDG11 on sustainable cities and communities, SDG 12 on responsible consumption and production, SDG 13 on climate action and SDG 17 on partnership for the goals.

The United Nations Centre for Trade Facilitation and Electronic Business (UN/CEFACT) has contributed to this by promoting the interoperability and harmonization of electronic data exchanges. It has specifically worked on the internet of things, smart containers and dematerialized bills of lading and other transport documents.[20]

Supply chain disruption and electronic cargo documents

The global pandemic, exemplified by COVID-19, has highlighted the vulnerabilities within supply chains worldwide. Amidst these challenges, electronic cargo documents have emerged as indispensable tools for risk mitigation. By digitizing documentation processes, businesses ensure seamless continuity in operations, even amidst lockdowns and travel restrictions. Electronic cargo documents provide remote access to crucial information, facilitating smooth goods clearance at borders while reducing reliance on physical paperwork (which is prone to delays and inaccessibility during crises).

Currently, traditional paper-based bill of lading (B/L) systems persist, necessitating physical couriering of documents: a process riddled with inefficiencies, expenses, and susceptible to fraud. Electronic bills of lading (eBLs) present a solution to mitigate such risks. By digitizing the B/L process, eBLs facilitate real-time transmission of documentation, granting stakeholders instant access to critical information throughout the shipment journey. This transparency and efficiency minimizes errors, delays, and disputes inherent in paper-based B/Ls, thereby enhancing operational efficiency and customer satisfaction.

Moreover, the adoption of eBLs aligns with broader industry endeavors towards digital transformation and sustainability. By eliminating the need for paper documentation, eBLs reduce paper consumption, curtail administrative costs, and diminish the

[20] See: https://unece.org/media/press/373109

environmental footprint of transport and logistics operations. Additionally, eBLs offer enhanced security features, such as digital signatures and encryption, fortifying protection against fraud and unauthorized alterations.[21]

The integration of digitalization and adherence to standards present significant opportunities to mitigate risks and bolster efficiency in transport and logistics operations. Leveraging electronic cargo documents and embracing eBLs, businesses adeptly navigate challenges posed by global pandemics and administrative inefficiencies, thereby ensuring the resilience and sustainability of supply chains in an increasingly digital and interconnected world.

These benefits aside, digital cargo documents are susceptible to various cybersecurity threats, including unauthorized access, posing significant risks to businesses within the supply chain. Without robust security measures in place, companies may face legal disputes, financial losses, and reputational damage. Thus, prioritizing stringent cybersecurity protocols, coupled with robust digital identity verification mechanisms, become imperative to protect sensitive information and uphold trust and credibility across the industry.

Regulatory compliance risks

Compliance with regulations governing transportation, customs, and trade is essential for supply chain operations. Failure to adhere to regulatory requirements can result in fines, penalties, and legal disputes, posing significant financial and reputational risks to businesses. For transport and logistics, the implementation of initiatives such as the EU regulation on Electronic Freight Transport Information (eFTI) or the European Maritime Single Window environment (EMSWe) brings forth several positive aspects. These initiatives streamline administrative procedures, reduce paperwork, and enhance data accuracy and transparency by standardizing electronic declarations and harmonizing reporting requirements. They also facilitate interoperability among stakeholders, contributing to operational efficiency and trade facilitation within Europe and beyond. However, one can argue that the transition to electronic information exchange systems also introduces certain risks. The reliance on digital platforms for transmitting sensitive freight and maritime information may raise concerns regarding cybersecurity vulnerabilities, data privacy, and potential system failures. There is a risk of unauthorized access, data breaches and disruptions to supply chain operations, highlighting the importance of robust cybersecurity measures and contingency plans. Despite these challenges, the overarching aim of these initiatives is to reduce overall risks and enhance the resilience, transparency and efficiency of transport and logistics operations, ultimately benefiting stakeholders across the supply chain.

Security threats

Security threats in the transport and logistics industry, including theft, piracy and terrorism, pose significant challenges that can lead to substantial losses. Addressing these threats requires a comprehensive approach that combines technological advancements with strategic measures.

Digitalization plays a crucial role in enhancing security by improving visibility, traceability, and control over shipments. Electronic documentation, such as eBLs and eFTI, streamlines administrative processes, reduces paperwork, and enhances data accuracy and transparency. Real-time monitoring of supply chain activities enabled by digitalization allows for timely intervention in case



[21] See: https://www.tradefinanceglobal.com/posts/dcsa-member-carriers-commitment-ebls-advance-trade-digitalisation/

of security threats or disruptions. By way of an example, this can be achieved through smart containers,[22] that integrate sensors for real-time tracking and monitoring, enhancing visibility, inventory management and security (they can detect unauthorized door openings).

Even these solutions can introduce some new, unique risks. The electronic components and batteries within such smart containers may pose ignition hazards, particularly for hazardous materials or goods prone to fire. Moreover, the connectivity of smart containers to digital networks exposes them to cyber threats such as hacking and data breaches. Conducting comprehensive risk assessments and implementing risk management strategies are crucial for identifying and mitigating security threats effectively. Robust cybersecurity measures should also be implemented to safeguard containerized data and prevent unauthorized access, ensuring compliance with safety protocols and regulations.

While smart containers and electronic documentation offer innovative solutions for optimizing supply chain visibility and efficiency, proactive risk management remains necessary to address the challenges and vulnerabilities associated with their deployment effectively. Adopting a holistic approach to risk mitigation and leveraging technology-driven solutions will enable businesses to harness the benefits of smart container technology and electronic documentation, while ensuring cargo security, operational integrity and regulatory compliance.

[22] See: https://unece.org/sites/default/files/2023-10/WhitePaper_ECE_TRADE_446E_SmartContainers.pdf



Building **risk competence**



Risk competence is defined as an ability or capability to manage risk (and uncertainty) within a work role. It is a complex term used to describe individuals, small groups, and whole organizations.

WP.6, through its publication *Risk Management in Regulatory Frameworks: Towards a Better Management of Risks*[23] argues that risk management should be a central process (involving all stakeholders such as regulators, policymakers, or a business) underlying all regulatory activity. Such stakeholders make decisions that require application of risk management tools (risk knowledge). The authors of the publication ask: "What is good risk management?". Their answer includes establishing the context and timely identification of risks, ensuring risks are properly analyzed and evaluated, and that a balanced risk treatment is chosen and efficiently implemented. However, merely completing such risk management activities does not guarantee successful, effective, or efficient risk management. Stakeholders and their agents must undertake such risk management activities with competence and their organizations must have the risk management capabilities required to ensure goals are met. Furthermore, risk management requires a collaborative approach within and across organizations.

Risk Management is the process whereby individuals or groups address risks associated with goals and activities. The necessity for risk management in regulatory systems is further highlighted in, ECE publication, *The Basics of Quality Infrastructure for Trade*.[24] Through market surveillance, Government agencies contribute to ensuring that products do not endanger health, safety, or any other aspect of public interest. This manifests in standards, technical regulations and enforcement plans based on an analysis of risk at all stages of the product lifecycle. The objective of risk management is to get to a tolerable level of risk in the most efficient way. Achieving this goal requires a formal risk management process, which will entail establishing the context, identifying the risks in a timely manner, evaluating the risks (most critical risks receiving the highest priority), choosing a risk treatment strategy, implementing that strategy and contingency planning (where the risk occurs). Applying best practice on risk management in regulatory frameworks and international trade results in regulations that make products safe, make organizations' processes stable and ensure consumers are better protected from hazards without stifling growth or creating barriers to trade.

For market surveillance authorities, properly identifying the risks that products placed on the market may cause is a prerequisite for developing timely and appropriate measures and ensuring marketplace safety, as per the ECE *Recommendation R on Managing Risks in Regulatory Frameworks*.[25] However, it is widely acknowledged that simply being aware of the risk management process, tools or broader risk theories does not in itself bring about the effective and efficient mitigation of risk. Rather, it is the activity and decisions of experienced people. They must know the right things to do and must do those things correctly. What is the knowledge and what are the skills, attitudes and the behaviors required by professionals seeking to mitigate risk in this context? How can organizations and individuals develop the capabilities to become risk competent?

[23] See: https://unece.org/info/Trade/WP.6/pub/2458

[24] See: https://unece.org/info/Trade/WP.6/pub/385732

[25] See: https://unece.org/fileadmin/DAM/trade/wp6/Recommendations/Recommendation_R_en.pdf

What does it mean to be risk competent?

A competence is a set of knowledge, skills, and attitudes (KSA) that manifest in our behaviors. They represent the underlying attributes that result in effective or superior job performance. Typically, behavioral proficiency indicators are described for each theme of the competence. Competency approaches are now an established integral feature of many private and public organizations. Risk competent professionals (individuals, groups of individuals, institutions, or organizations) can manage risk and uncertainty within their role. The European Entrepreneurship Competence Framework (EntreComp) is a common reference framework that identifies fifteen competences,[26] one of which is "**coping with uncertainty, ambiguity and risk**." Whereas the EntreComp framework is targeted at entrepreneurs, there are other risk-focused competency frameworks with broader target audiences.[27] The Institute of Risk Management (IRM) publish a framework[28] for those working in risk management and describes what risk professionals should know and be able to do. The IRM advocates that managers should ensure risk management is incorporated into their thinking, planning, organizing, and controlling. Risk competent managers must understand the internal and external environment of an organization and its implications for risk management. Risk competent managers may also need to be able to identify, analyze and evaluate the nature and impact of risks and opportunities in their business area. Furthermore, they may be expected to develop, select, and implement risk treatment strategies and controls. In respect of people management, the risk competent manager is likely to influence the risk behavior of others, support individuals and teams in the practice of risk management and develop risk competence in the workforce and workplace. Thus, risk competence generally reflects abilities to identify, analyze, assess and prioritize, treat or control risks within accepted levels of tolerance and risk attitude and in a cost-effective manner. A risk competent professional is one who exemplifies (relevant) behaviors such as:



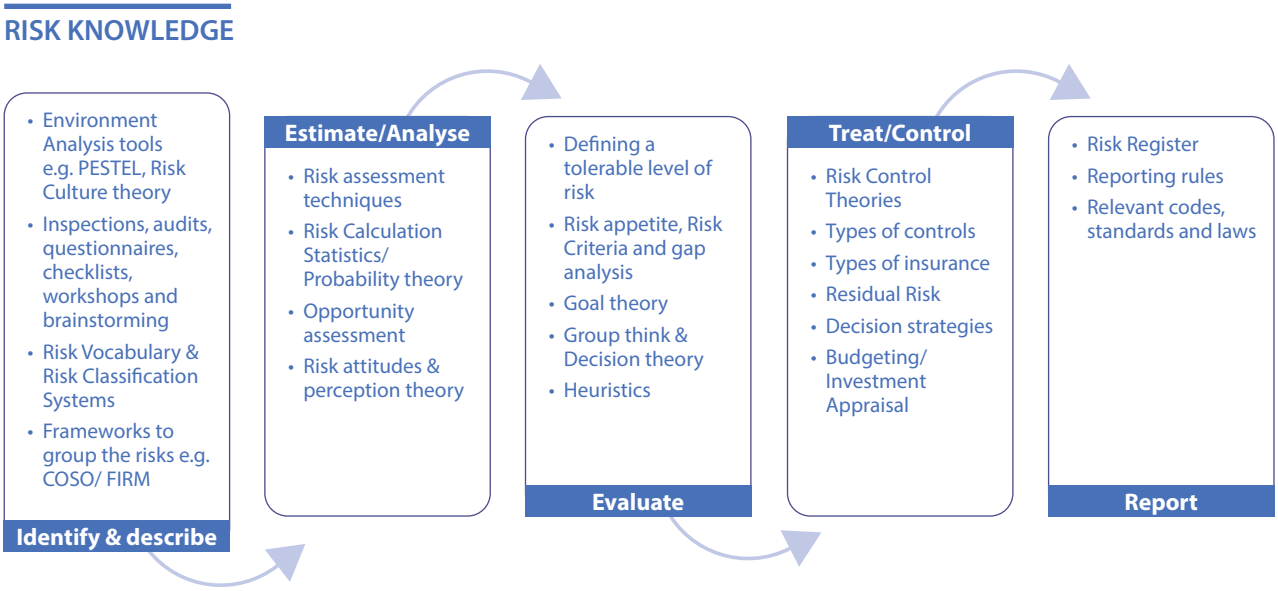
- Showing awareness of how the way work is undertaken might affect risk elsewhere; i.e., how actions and decisions may introduce threat(s) or create vulnerability.
- Applying relevant risk-related knowledge, skills, tools, techniques, standards, and frameworks in the pursuit of work-related goals.
- Ensuring regularly and in a timely manner that risks faced are identified, analyzed, assessed, and prioritized (and if necessary, reported).
- Taking proactive steps to reduce threat and maximize opportunity to attain goals within work constraints, by evaluating risk and treating intolerable risks in an effective and efficient manner.
- Ensuring resilience (adaptive capacity, considering continuity and contingency).

[26] European Commission, Joint Research Centre, Bacigalupo, M., Kampylis, P., Punie, Y. *et al.*, *EntreComp – The entrepreneurship competence framework*, Publications Office, 2016, https://data.europa.eu/doi/10.2791/160811

[27] Kelly, P. (2023), "Are you competent coping with uncertainty and risk? Implications for work-applied management", Journal of Work-Applied Management, Vol. 15 No. 2, pp. 241-251. https://doi.org/10.1108/JWAM-10-2022-0071

[28] IRM (2002), "A Risk Management Standard", AIRMIC, ALARM, IRM, pp. 1–14. Available from https://www.theirm.org/media/4709/arms_2002_irm.pdf (accessed 12 Feb. 2024)

FIGURE 3: KNOWLEDGE COMPONENTS OF RISK COMPETENCE



Source: WP.6

How can you develop risk competency?

It is important to recognize that risk and risk competence are associated with context (a variable). Context will include aspects of the organization, its (risk) culture, the national culture (values such as uncertainty avoidance or tolerance for ambiguity), specific goals and role. Individuals and organizations also vary in risk attitude and appetite, their risk tolerance (readiness to bear the risk after risk treatment) and the way they seek to treat and control risk. They also differ in their use of risk strategies such as risk avoidance, risk sharing (distribution of risk with other parties) and extent of risk financing. Consequently, individuals and organizations vary in the KSAs and behaviors they require to be risk competent – KSAs underpin risk competence and associated risk behaviors. Risk knowledge is extensive.[29] For example, there is a vast number of academic journal, magazine and news articles containing "risk" in their title, with the addition of textbooks, standards and on-line resources; this creates a wealth of risk knowledge for practitioners to embrace. We can use the risk management process (the core set of risk activities) and its outputs as a reference point to map specific risk knowledge requirements, see figure 3.

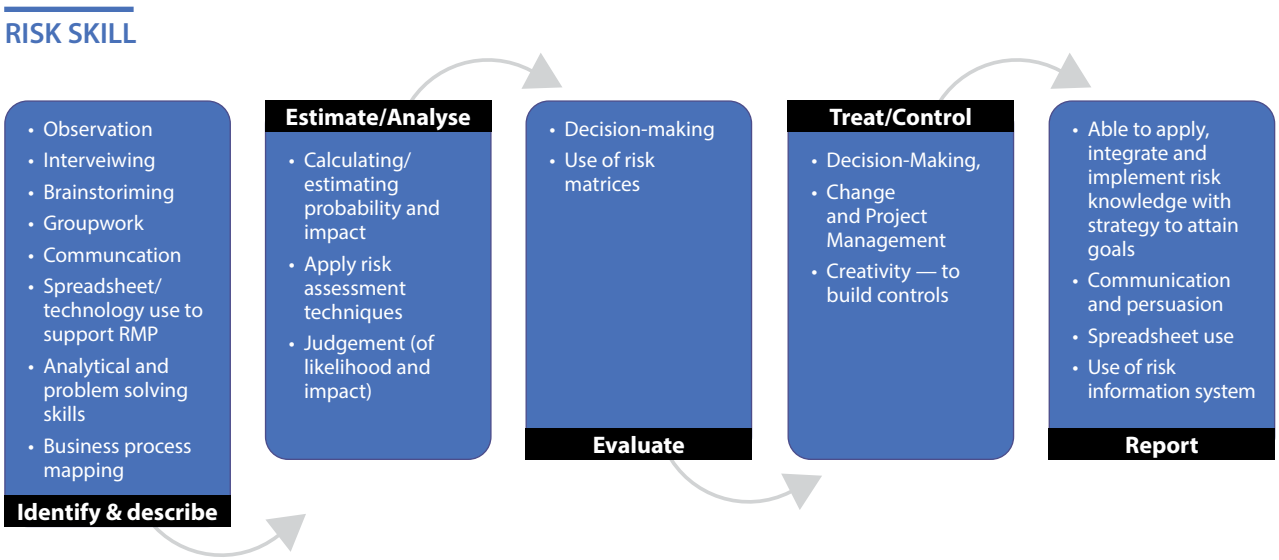
Risk knowledge has various attributes such as level and depth. At the base level (understanding), a professional may simply be aware of the risks that impact upon goal attainment and be able to contribute to risk decisions. At the next level they may be able to analyze (identifying and assessing) risk and apply limited areas of knowledge, e.g., to select appropriate risk controls. In the most advanced level, the risk-competent professional should be able to evaluate risk and be able to rate, prioritize and justify actions. Furthermore, they may be able to create risk tools (such as risk checklists or registers) or design risk controls. There are numerous sources of risk knowledge.[30]
The body of risk knowledge refers to the core teachings and skills required to become risk competent. There can be much to learn about risk identification, analysis and assessment. Initially, this may derive from general risk textbooks. Once analyzed within the risk management process, the next step will be to consider and make decisions about what and how to control a particular risk. As a starting point, understanding generic aspects of management control will help towards treating or controlling risk. Risk knowledge can also be gained from a variety of risk standards[31] and publications from organizations such as the Committee of Sponsoring Organizations of the Treadway Commission (COSO). The risk competent professional may also be aware of relevant codes and legislation such as the Sarbanes-Oxley act of 2002 (SOX) or the UK corporate governance code. Finally, there are a variety of organizations such as the IRM that offer risk training to exchange risk knowledge.

[29] See: Kelly P. (2018), "The Evolution of Risk Management Thinking in Organizations", Engemann K. (Ed.), The Routledge Companion to Risk, Crisis and Security in Business, Routledge, London and New York, pp 20-46.

[30] See, for example ECE *Risk Management in Regulatory Frameworks*, op.cit.

[31] See for example ISO 31000 series of standards.

FIGURE 4: SKILL COMPONENTS OF RISK COMPETENCE



Source: WP.6

Aside from growing their risk knowledge, risk competent professionals will gain experience in application, developing risk skills. The ability to use risk knowledge effectively leads to the development of risk skill – the ability to perform risk-related tasks and activities well.

Risk skills may be categorized as generic management skills or may be specific to risk management. They may be regarded as technical (based on risk knowledge application) or people skills (communication, relationship, analytical and management). Analyzing and estimating risk is often a key skill of the risk competent manager as risk decision making quality depends upon it. The ability to calculate risk has become increasingly difficult and error prone. When operating in turbulent environments, and in the absence of historic risk-related data, managers often need to make subjective judgments about risk; this is a skill born out of experience. Risk estimation and assessment is a precursor to response, control, and treatment. The range of skills required can be extensive, and dependent upon role and context. A selection of risk skills, mapped against risk management process tasks and activities, is indicated in figure 4.



The risk management technical skills of the competent risk professional are likely to be broader and more honed than those of others with a general interest in risk. There is likely to be a need for skills associated with planning (risk management strategy), implementing a risk management architecture (change management); measuring risk management performance and learning from collective risk management experience. Examination of the vast range of risk-related knowledge and skills emphasizes the need for individuals to tailor their own

definition of risk competence and own framework of knowledge, skills, and behavioral indicators for their role needs. Many aspects of risk competency are encapsulated in the ability to create or contribute to a risk register, a record of information about identified risks. This is a key output of the application of risk related KSAs. The register should identify risks faced, controls currently in place, additional controls required and responsibility for control activities. The register is dynamically updated as different activities in the risk-management process are completed.



Part 3:

Emerging **issues**

The role of risk-based regulations in **advancing the transition towards circular economy**



Risk-based regulations are fundamental for ensuring a successful transition to a circular economy, understood as one that replaces production with sufficiency: reuse what you can, recycle what cannot be reused, repair what is broken, remanufacture what cannot be repaired, transforming goods that are at the end of their service life into resources for others, closing loops in industrial ecosystems and minimizing waste.

By virtue of integrating a systemic approach for the identification, assessment, and treatment of both foreseeable and unforeseen events throughout decision-making processes, risk-based regulations provide governments with the required flexibility to adapt to changes in market conditions. In particular, the advancements in production technologies, materials recovery methods and business models, which are continuously evolving to respond to the triple planetary crisis (of climate change, pollution and biodiversity loss), digital transformations, and other cross-cutting agendas. This flexibility ensures proactive governance while preventing over-regulation, ensuring that resources are allocated in a cost-effective manner.

Further, risk-based regulations are typically better received by industries, as they are seen as more rational. This not only improves regulatory compliance but also encourages innovation and breakthroughs that further the principles of the circular economy and the consolidation of new transformative development pathways for strengthening resilience.

Several countries have developed risk-based regulations for enabling a successful transition towards a circular economy. While their experiences vary, they all exhibit structured approaches that are in line with internationally recognized principles and practices (for example, those outlined in the ISO 31000 standard on risk management). This includes tackling higher risk areas and referencing standards in technical regulations for guiding implementation and enforcement. Below are three case studies, demonstrating the integration of risk-based regulations for circular economy into national and regional regulatory frameworks.

Case study: construction materials regulations

Established to enhance the sustainability of buildings, the *Green Building Standards Code of California* comprises risk-based technical regulations that consider the lifecycle impacts of buildings on the environment. The *Code* applies to all types of new buildings, including residential and non-residential as well as additions and alterations that increase a building's conditioned area, volume, or size, and is aimed at promoting, among others:

- **Energy efficiency** through energy performance requirements, including improved insulation, high-efficiency windows, and lighting systems. The *Code* also requires the use of renewable energy sources like solar panels.
- **Water efficiency** through strict water efficiency requirements for plumbing fixtures, including low-flow toilets, showerheads, and faucets. The *Code* also requires the use of drought-tolerant landscaping to minimize the need for irrigation.
- **Design for reuse** through promoting the efficient use of materials and resources during construction and throughout the building's lifecycle. The *Code* requires designing for future adaptability, potential disassembly, and reuse of materials.
- **Recycling and reuse** of materials through mandating that a minimum percentage of materials used in construction be sourced from recycled or recyclable

materials. The *Code* also requires that construction projects divert a percentage of the debris generated away from landfills to recycling facilities.

- **Indoor environmental quality** through provisions to improve indoor air quality, such as using low-volatile organic compounds materials for paint, flooring and adhesives. The *Code* also sets requirements for ventilation and moisture control, all aimed at creating a safer and more comfortable indoor environment for occupants.

Case study: single-use plastics ban

Canada's *Single-use Plastics Prohibition Regulations* (SUPPR) form a critical element of the Government's strategy to combat pollution and achieve zero plastic waste by 2030, while contributing to the reduction of greenhouse gas emissions. It comprises risk-based regulations, which ban the production, import, and sale of the certain single-use plastic items that are commonly found in the environment, are often not recycled and have readily available alternatives. These items include:

- Plastic check-out bags
- Cutlery (e.g., forks, knives, spoons, chopsticks)
- Foodservice ware made from or containing problematic plastics, including containers and lids used for food prepared or packed on premises
- Stir sticks
- Straws (with certain exceptions)
- Six-pack rings or ring carriers used for beverage cans and bottles

To ensure accessibility for individuals with medical needs, SUPPR provides an exception for single-use plastic flexible straws, denoting that these will be available for purchase in packages of 20 or more from retailers and can be provided upon request at restaurants and other establishments. Further, implementation follows a phased approach to provide enterprises time to adjust their practices and deplete existing stocks. The Government also committed to supporting these regulatory efforts with actions aimed at improving national recycling systems and encouraging the development and adoption of sustainable alternatives to single-use plastics.



Case study: extended producer responsibility

The EU is implementing several extended producer responsibility (EPR) schemes, with the aim of bolstering efforts for addressing the risks associated with post-consumer waste. Launched under the revised EU *Waste Framework Directive*, the schemes address these risks at the source by requiring producers to take responsibility for the collection, recycling, and disposal of products once they are no longer usable, ensuring that resources are used more efficiently and sustainably.

The EPR schemes target a range of products, including packaging, electronic and electrical equipment, batteries, vehicles and textiles, with implementation following a phased approach to allow enterprises to build the required capacities for compliance. Each product scheme has specific targets and regulations tailored to the particular waste management challenges facing the industry. Key elements of these schemes include the following.

- **Producer responsibility:** EPR schemes shift the responsibility for end-of-life product management from municipalities and consumers to producers. This ensure that producers consider end-of-life management as part of their product design.
- **Design for environmental sustainability:** EPR schemes encourage manufacturers to design products with the environment in mind. This includes using materials that are easier to recycle, reuse and repair; reducing the use of hazardous substances; and, designing for longer product lifespans.
- **Financial responsibility:** Producers are financially responsible for the costs associated with the collection, treatment, and recycling of their products. This may involve paying fees to a collective scheme that manages these activities on behalf of multiple producers or taking direct action to manage their products' waste streams.

- **Reporting:** Producers under EPR schemes are required to provide evidence of compliance with recycling and recovery targets and associated EU harmonized standards. This includes detailed reporting on the amounts and types of materials collected and recycled.



Challenges to implementing risk-based technical regulations for the circular economy

Governments face several challenges in implementing risk-based technical regulations for the circular economy. As is the case of any technical regulation, the complexity of the associated risks, often interrelated and requiring a multifaceted response, clashes with the limited, area-focused mandates of government agencies. The lack of scientific data is another challenge and is complicated by the difficulties in engaging all the relevant stakeholders, given the subjective nature of risk perception among regulators and the public as well as the public's expectation for absolute risk elimination.

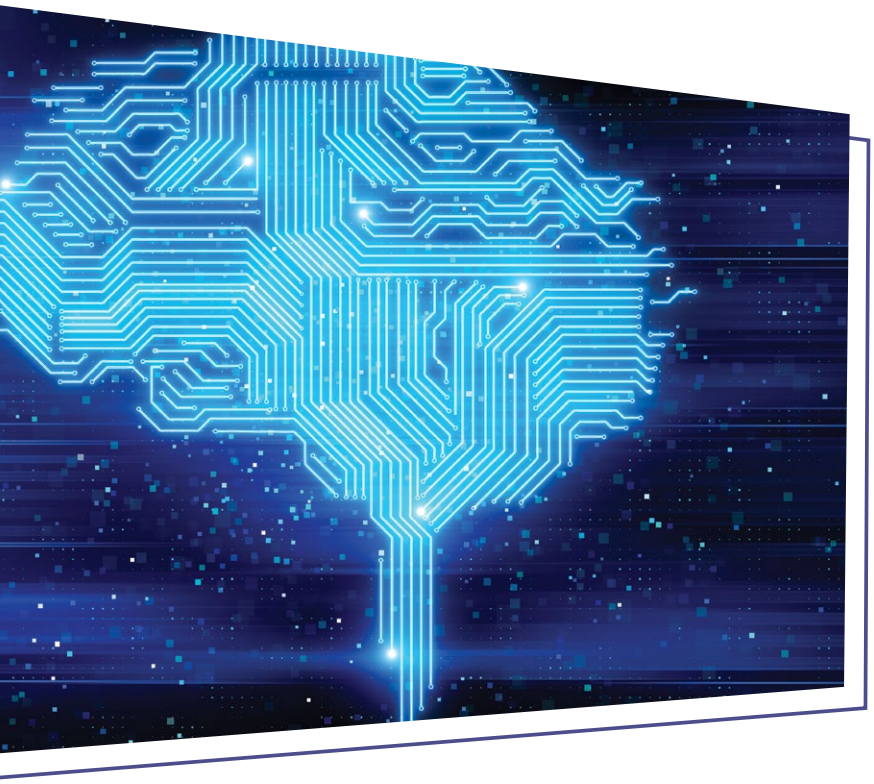
At the same time, in many countries, the definitions of the circular economy within many of the existing laws are often broad and lack specificity regarding the boundaries and scale of circularity. For instance, some laws do not specify whether a single instance of reuse, recycling, or recovery is sufficient to qualify as circular. Those stipulating multiple actions, do not set the exact number of recycling or reuse loops necessary to truly constitute a circular economy system. For the concept of circular economy to be impactful and practical, it is crucial to address these aspects clearly, ensuring the minimization of waste and efficient use of resources across both technical (abiotic) and biological production cycles.

For many developing countries, the above challenges are compounded by weaknesses in inter-agency coordination and slow legislative processes (often due to recurrent government restructuring). Further, many exhibit capacity shortfalls within line ministries and quality infrastructure agencies (standardization agencies, conformity assessment bodies and agencies dealing with metrology), making it difficult to ensure proper monitoring and successful implementation.

At the same time, implementing circular economy-related technical regulations involves significant challenges for enterprises in both developed and developing countries, particularly for small and medium sized-enterprises. These include, among others, the administrative complexities associated with reporting requirements, the lack of resources for financing investments in modern technologies and production methods, and variations in national regulations that complicate compliance for enterprises operating in multiple jurisdictions. In addition, consumer engagement, which is critical for the transition to circularity, is often limited by a lack of awareness or the inconvenience of returning products for recycling. Many countries also lack a viable market for recycled materials, which is critical for transitioning to circularity.

To help countries address these challenges, WP6 has revised *Recommendation T on Standards and Regulations for Sustainable Development*, which provides a practical approach for formulating robust risk-based technical regulations using risk management tools – including standards. The new edition emphasizes that risk-based regulation design and delivery must be subject to continuous review, which should be conducted on a cross-sectoral level to take into account changes in the relationship among risks and risk treatment strategies as well as new risks.

Evaluating **residual risk of AI systems**: Ensuring safety and regulatory compliance



Emerging regulatory frameworks on artificial intelligence (AI) require AI systems to have an acceptable level of residual risk. The concept of acceptable (or tolerable) risk as a prerequisite for a regulatory approval of a product is not new; its implementation in the field of AI, however, is associated with many challenges that need to be addressed in order to ensure safety of AI systems without hampering their development.

Residual risk in AI regulation

Regulating AI systems is a broad, multifaceted challenge for authorities all over the world. The concept of residual risk is becoming a horizontal issue in the context of AI regulation and compliance in all fields of application. Any regulated aspect of an AI system, i.e., any hazard that must be addressed – be it ethical concerns such as gender discrimination or technical issues such as cybersecurity – requires a demonstration that the level of the residual risk is acceptable.

The *EU AI Act* is one example of how the concept of residual risk is used in a general AI regulation. It requires "the risk management measures [of an AI system provider to] be such that any residual risk associated with each hazard as well as the overall residual risk of the high-risk AI system is judged acceptable, provided that the high-risk AI system is used in accordance with its intended purpose or under conditions of reasonably foreseeable misuse"[32].

The *EU AI Act* also requires that "those residual risks" be communicated to the user. A list of AI systems deemed to be high risk is provided in the Annex to the regulation; the term "high-risk" in this context implies potential serious consequences if a system is non-compliant. This could take the form of misbehavior or failure. It does not take into account the probability of such an event.

Compliance with standards as a way of demonstrating that the risk is tolerable

The concept of tolerable level of risk as a condition for regulatory approval is not new. Regulatory frameworks, especially in the field of technical regulation, aim at ensuring safety without stifling growth. Absolute safety cannot be a valid regulatory objective; there is no other way but to determine safety objectives in terms of a tolerable level of risk. ECE *Recommendation R on Managing Risk in Regulatory Frameworks*[33] was one of the first international best practice documents to present regulation as a risk mitigation tool which recommended that "regulatory authorities […] establish, implement and maintain, a process

[32] See: *EU AI Act, Final Draft* Article 9.4. https://artificialintelligenceact.eu/the-act/ Formally adopted by the European Council on 21 May 2024; not yet entered into force at time of writing.

[33] See: https://unece.org/fileadmin/DAM/trade/wp6/Recommendations/Recommendation_R_en.pdf

for determining, analyzing, reviewing and monitoring an acceptable level of risk within a regulatory framework".

Products that do not have an integrated AI or other digital technology often do not require the regulatory authority or economic operator to develop tools and perform evaluation of the level of risk associated with the product in order to establish regulatory requirements or to get a regulatory approval. In traditional regulatory frameworks, products that comply with regulatory requirements (which, in most cases means that they comply with the standards referenced in the regulations), are deemed to have a tolerable level of risk and thus are considered safe. So, in practice, to make sure that only products with a tolerable level of risk can be placed on the market, a regulatory authority needs to identify relevant standards, and an economic operator has to demonstrate compliance with them as a proof that the product is safe. A toy producer, for example, to ensure that the risk of the event in which e.g., a child swallows a battery that fell out of the toy, is tolerable, can demonstrate compliance with applicable standards, that contain requirements on materials and other characteristics of toys aiming to minimize the probability of such an event.

AI systems require different approaches for showing the acceptability of risk

This approach of compliance with standards as a way of demonstrating the acceptability of the product risk works well when the level of risk of a product depends on product characteristics that are known, or can be determined, measured and hence regulated, and when it is known how different product characteristics result in different levels of risk (as in the example of a toy). If this is the case, a regulation can describe the regulated product, and conformity with regulatory requirements can be established by testing the product and making sure it meets the requirements of the regulation or of a relevant standard. Which standard to apply and "how safe is safe enough" is still a challenging question for any regulatory authority, but once a standard is chosen, assessment of conformity can be rather straightforward.

This is not the case with AI systems, as they are by their nature grey or black boxes, meaning that it is impossible to look inside and know exactly what is contained or how it will react in all scenarios. They rely on uninterpretable algorithms, the functionality of which is only partly known or unknown. Not only can the functionality of an AI system not be exhaustively described in a standard, even if this were the case an economic operator or a conformity assessment body cannot easily look inside these systems to test them. Also, their functioning can be stochastic, meaning that they can behave differently in similar conditions.

This is why regulatory frameworks for AI systems require different approaches for ensuring that the level of risk of a system is tolerable. Indeed, emerging regulations in the field of AI are often more risk-based than traditional regulatory systems. They (i) establish requirements for AI system providers/other stakeholders to mitigate the risks of a system during its development (most commonly regulations also set out risk management processes and mitigation methods that must be applied to a system) and (ii) require the residual risk of an AI system to be acceptable.

To demonstrate that an AI system has been developed in conditions that are intended to mitigate its risks and is achievable by the means of usual conformity assessment procedures requires a system provider or a conformity assessment body to evaluate conditions in which the system has been developed. International standards on risk management systems for AI (such as ISO/IEC 42001) are being developed; demonstrating conformity with these





standards may not be very different from usual conformity assessment procedures, such as management system certification. Importantly, these standards are not "product standards" – they do not determine the safety of an AI system throughout its entire lifecycle, and conformity with these standards alone may not be sufficient for the system to be considered safe.

The second condition – acceptability of residual risk – thus becomes the main criterion on which regulatory stakeholders can rely in making judgments on whether the system is safe. Evaluation of residual risk poses a significant challenge and requires the development of new tools for testing of AI systems.

What is residual risk?

Residual risk is not strictly defined in many safety regulations. Historically, the logic of using residual risk has been in the context of introducing controls aimed at risk mitigation (residual risk is also a tool for evaluating the effectiveness of such controls). For example, a company could evaluate the risk of a computer virus disrupting its systems and decide to buy a certain anti-virus software; the risk of the computer virus disrupting systems after the antivirus software has been installed is the residual risk.

Requirements for a general framework for evaluation of residual risk

Residual risk should be demonstrated in the same format as the risk before the mitigation measures have been introduced. Building a framework for the evaluation of residual risk requires the identification of the format in which the risks of an AI system should be presented, so the methodology for identifying the residual and non-residual (often referred to as "inherent") risk of an AI system will be similar.

To evaluate the risk of an AI system, we need to evaluate all the situations in which the system will fail and cause harm (against all hazards), as well as how likely these scenarios are.

As it may prove complex to look inside an AI system to identify the hazards that can materialize, as well as the likelihood and level of harm of risk events, the only possible way to evaluate the risk of an AI system may be to rely on the results of its functioning in any given situation. Ideally, to evaluate the risk, an AI system should be run through all the scenarios it can possibly face in a simulation (so that no real harm is caused). Results of its functioning can be turned into an evaluation of its risk; if a number of scenarios are identified in which the system has caused harm, the probability of these scenarios together with the level of harm can be sufficient to make a decision on whether the system is safe enough.

The main challenge with this approach is that in most cases it is impossible to run an AI system through all the scenarios it can possibly face. Moreover, it is impossible to know all such scenarios. AI systems are complex systems that operate in complex environments, and there is an infinity of scenarios they might face. There is no other way but to limit the number of scenarios, meaning that the results of the risk evaluation will be dependent on how scenarios are chosen.

Any framework for assessing the residual risk should determine:

- How scenarios are chosen, so that most likely and most dangerous scenarios are covered

- How the behavior of an AI system in each scenario is evaluated against the hazards and turned into a set of metrics that can be used for risk evaluation
- How to identify a testing threshold which ensures that enough testing has been performed

Addressing these challenges requires building a framework that would ensure that the evaluation of residual risk encompasses dangerous scenarios that were identified in testing, as well as the potential harm that can occur in scenarios that were left untested. Such a framework should include at least the following processes:

- **Identifying** all possible hazards and risk events that could materialize during the functioning of an AI system that could cause harm
- **Building** a list of situations/ scenarios that a system can face
- **Identifying** which hazards can occur in each scenario
- **Evaluating** potential severity of hazards in scenarios and their frequencies
- **Selecting** scenarios for testing based on the level of risk, ensuring coverage of the most probable and most dangerous scenarios
- **Performing** simulation/test and evaluating the residual risk

At the time of writing, there are no fully-completed methodologies that can be used for the evaluation of the residual risk of AI systems. There are fields – such as validation of ADS – in which these tools have been developed and some of them can be generalized and replicated to other fields.

Conclusion

Evaluation of residual risk will be an indispensable part of the conformity assessment process of all regulated AI systems. No matter if performed by a system developer or by a third-party conformity assessment body, addressing the challenges related to the evaluation of residual risk described in the previous sections necessitates application of scenario-based approaches and simulation methods within regulatory frameworks. These methods are already being developed in several industries facing the challenges of regulatory approval of black/grey boxes.

Developing a comprehensive framework for evaluation of the residual risk of AI systems is essential for ensuring safety and facilitating trade, and recommendations developed by WP.6 GRM (especially R and S) can be used in the development of the required methodologies and tools. On a more general level, the GRM can be a platform for international cooperation in the field and the advancement of these techniques.

Gender considerations for risk management in regulatory systems

While risk management may appear to be a neutral mechanism, it reflects the expertise and lived experiences of those conducting the assessments, and the limitations of the tools they use. As this document highlights, standards, technical regulations and conformity assessment, among other instruments, can be effective tools to help safeguard against hazards. However, as emphasized in the ECE's work on gender-responsive standards, these tools often have hidden bias imbedded within them.

The ECE *Guidelines on Developing Gender-Responsive Standards* encourages developers and users of standards to reflect on who is at the table during the development process. The *Guidelines* highlight how difficult it can be to identify solutions to problems if you are not aware the problem exists. For example, smartphones, which have continuously become physically larger over the years, are often too large for women's hands and pockets.[34] This makes the devices more precarious for women, or users with small hands, to operate these devices. If the smartphone design team is made up entirely of men, they may fail to recognize that they are designing a product that can only be used comfortably, and safely, by half of the population.



Women are under-represented in standards development. Standards and accreditation bodies who signed the ECE *Declaration on Gender-Responsive Standards and Standards Development*,[35] are called upon to increase the participation of women in the development of standards. But change takes time. As standards bodies begin to include more women and incorporate gender-responsiveness in their standards, regulations may still lag behind, given how long standards take to develop and how slow regulatory systems are to change. Consequently, it is important for those involved in the risk management process to consider how biases may inadvertently influence the resources they choose and/or the assumptions they make.

It is becoming increasingly well documented that one of the root causes of hidden biases is due to the pervasive exclusion of women. For instance, women have historically been excluded from medical studies.[36] The gender gap in the medical field has a multitude of implications: misdiagnosis, incorrect dosage, and dismissing critical patient information, to name a few. This is important in the context of risk management because we often rely on scientific studies to help shape standards, regulations, or policy direction. The TBT Agreement states: "…technical regulations shall not be more trade-restrictive than necessary to fulfil a legitimate objective, taking account

[34] Perez, Caroline Criado (2019) "Invisible Women: Exposing Data Bias in a World designed for Men," p. 159.

[35] https://unece.org/trade/wp6/Gender-Resp-Stdards-declaration

[36] Merone, Lea, Tsey, Komla, Russel, Darren, and Nagle, Cate (2022) "Sex Inequalities in Medical Research: A Systematic Scoping Review of the Literature," Women's Health Report.

of the risks non-fulfilment would create. […] In assessing such risk, relevant elements of consideration are, *inter alia*: available scientific and technical information…"[37] The reliance on scientific information presumes a certain level of rigor, but fails to recognize the implicit gender bias that may exist within said information. As such, the TBT agreement is one example of hidden bias that is proliferated through a reliance on scientific research.

Indeed, gender-responsiveness is an essential consideration for risk management. The risk management process provides a window of opportunity to take stock of the policy or technical objective at hand and reflect on the path forward. At the heart of risk management is the goal of protecting consumers from hazards. Research has demonstrated that standards are not currently protecting women as well as they protect men.[38] Therefore, applying a gender-responsive lens throughout the risk management process is crucial. This is particularly important when looking at risk management as part of the regulatory process. Regulations are typically developed to protect health, safety, security, and the environment. A diversity of perspectives in the risk management process can help ensure regulations, and by extension the products and processes they regulate, are safer for everyone.

The good news is that, as the concept of gender-responsiveness becomes more salient, more resources are being developed. WP.6 resources, such as the *Guidelines on Developing Gender-Responsive Standards* and *Why Gender-Responsive Standards are Better for Everyone,*[39] provide helpful advice on how to begin incorporating a gender lens in various areas of work. In Canada, the Cabinet Directive on Regulation (the instrument that dictates how federal regulations should be developed, managed, and reviewed) includes an obligation for regulators to conduct a gender-based analysis plus (GBA+). The GBA+ is an analytical tool used to help identify the impacts of government initiatives on diverse groups of women, men, and gender-diverse people. The "plus" acknowledges that GBA goes beyond sex and gender differences, to consider multiple identity factors that intersect to make people who they are (such as age, disability, education, ethnicity, economic status, geography (including rurality), language, race, religion, and sexual orientation).[40] The Government of Canada maintains a publicly-available website with a plethora of tools, videos, and trainings. These resources can help inform a risk management process that goes beyond gender-responsiveness.

Often, we find ourselves looking for simple answers to complex problems. Knowing how and where to start can be overwhelming. However, there is one recommendation that comes up time and time again: to look around the table to see who is helping to develop your risk assessment. If everyone looks the same as you, then there's a good chance you've missed something.

[37] World Trade Organization, *Technical Barriers to Trade Agreement*, Article 2.2.

[38] See: Parkouda, Michelle (2020) *When one size does not protect all: Understand why gender matters for standardization*. Standards Council of Canada: Ottawa.

[39] See: https://unece.org/info/Trade/WP.6/pub/381789

[40] See: Government of Canada, Cabinet Directive on Regulation, Article 5.2.3. https://www.canada.ca/en/government/system/laws/developing-improving-federal-regulations/requirements-developing-managing-reviewing-regulations/guidelines-tools/cabinet-directive-regulation.html

New Zealand risk engine

This article describes the regulatory system applied in New Zealand for the regulation of electrical and electronic consumer products. New Zealand and Australia operate a single economic market with slightly different controls, but full mutual recognition.

New Zealand's electrical and electronic product regulatory system

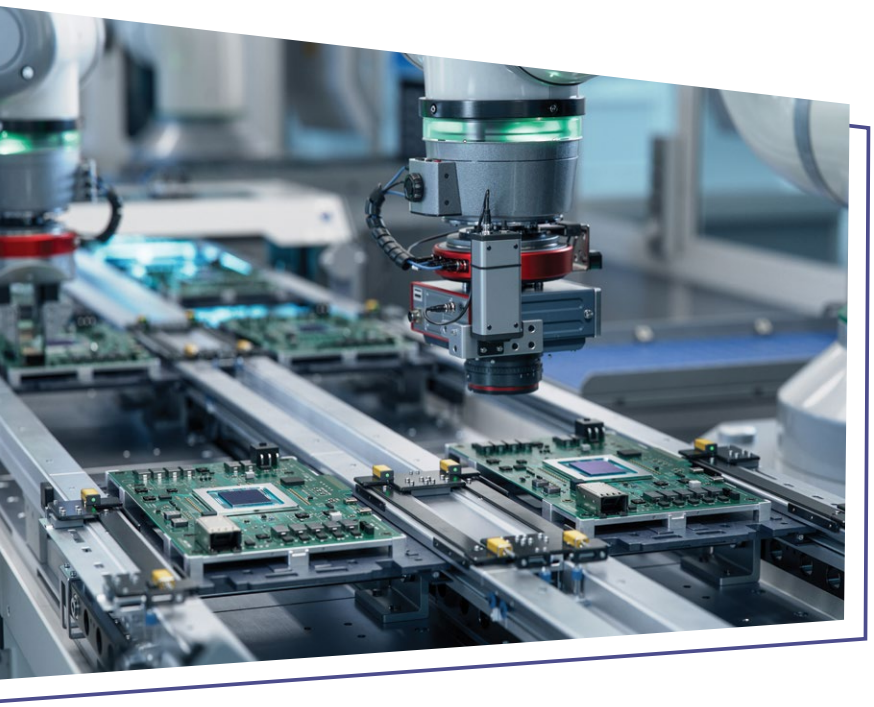
Good regulatory practice and WTO conventions expect that regulatory interventions are proportional to risk, and that international standards are used as the basis for assessing compliance with regulatory objectives.

New Zealand operates an electrical and electronic product regulatory system that implements three risk levels of regulatory pre-market intervention guided by assessed product safety risks. When New Zealand was looking into the number of risk levels applied to this regulatory intervention, it was concluded that a form of regulatory intervention with three risk levels would be sufficient and appropriate for the New Zealand regulatory system. This was supported by the outcomes of industry and consumer consultation.

Regulatory interventions

For all products, essential safety is required, and applicable standards that are deemed to be safe are provided in the regulations.

- **For high-risk products**, an approval by the New Zealand regulator or a recognized independent product certification is required. These are established under New Zealand's mutual recognition arrangements.
- **For medium-risk products**, a supplier declaration of conformity is required to have been issued by the supplier and available to the purchaser. A supporting test report issued under a relevant standard has to be available for regulatory market surveillance.
- **For the products felt outside** of high risk and medium risk categories, the essential safety intervention applies as a light-handed regulatory intervention.



To support the regulatory system, New Zealand developed a risk assessment tool (a risk engine) to provide guidance on the appropriate level of regulatory intervention that should be applied. The regulatory system of the majority of Australian State and Territory regulatory agencies is almost identical to the system operated in New Zealand, and Australia electrical regulators have a similar type of risk assessment tool used to assess the risk level for electrical products.

The development of New Zealand's analytical risk assessment tool began in 2006. The design was based on the consideration of more than ten years of product non-compliance-related incidents in New Zealand and Australia.

When the risk engine was being developed, further collaboration with the regulatory agencies of the Association of Southeast Asian Nations (ASEAN) and the Chinese regulator for the China Compulsory Product Certification (CCC) system were conducted.

New Zealand risk engine

The New Zealand risk engine determines and separates each individual product assessed into an appropriate risk level. The output of the risk engine also provides guidance on which regulatory intervention should be applied to that particular product.

The risk engine was designed based on applying the principles that risk is a combination of consequence and probability. **Risk = Consequence x Probability (R= P x C)**. "P" represents the probability factors (P) for the probability of non-compliance. "C" represents the factors for consequence, also referred to as technical factors (T) to represent factors for the consequence of technical safety hazard.

The risk engine assesses the risk of non-compliance

The probability of non-compliance is driven by a series of factors, most of which relate to market conditions, and can be assessed. This probability of non-compliance with the safety standards is used as the probability factor "P" for the engine. Each individual probability factor has been identified as a potential cause of non-compliance for a particular product, such as the cost of testing, the availability of testing facilities, the availability of international standards and the use of new technologies.

Regulatory intervention is applied to offset the level of risk created by using a product that is present without that regulatory intervention. It is implemented through changing the probability of non-compliance. For example, where a standard is available to be applied, the regulatory intervention would normally be focused on influencing compliance with that standard.

A series of technical (consequence) factors "T" that represent the consequence of technical safety hazard have been built into the risk engine, based on the specific dangers that the particular product creates, such as the use of the product by children, or its common usage being in damp conditions.

Ten probability factors and twenty-four technical factors have been developed and built into the risk engine.

Examples of probability factors:

- Product uses new technology or lacks a relevant international standard
- Product is controlled internationally using standards considered inadequate for local application
- There are cost disincentive for compliance
- Standards recently updated
- Compliance with the applicable standard is complex or technically difficult



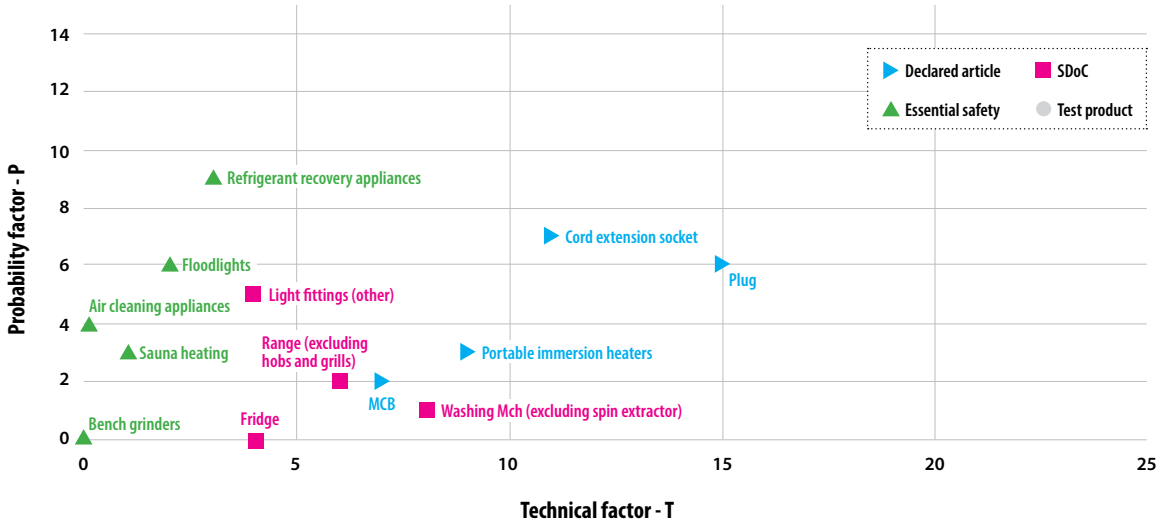
Examples of technical factors:

- Product providing a safety functions
- Product likely to move during or between uses
- Product likely to be used by unsupervised or lightly supervised children
- Product subject to likely significant misuse
- Product uses toxic substances
- Product relying on guards and barriers to prevent mechanical injury
- Product used in circumstances where the user is not able to readily disconnect with normal physical reaction to electric shock
- Product has high-energy storage
- Product relies on isolation between low voltage (LV) and exposed extra low voltage (ELV) parts
- Product's standard is recognised as being barely adequate.

Each factor is given an equal rating. The probability and technical (consequence) rating is simply the number of factors that apply to a particular product.

More than 200 electrical and electronic products have been selected and their data fed into the risk engine for assessment. The output of the assessment from the risk engine is depicted in a graphical representation. The probability factors and technical (consequence) factors are plotted on two axes. Below is an example of the graphical representation of the risk engine.

FIGURE 5: EXAMPLE: NEW ZEALAND RISK ENGINE



Source: New Zealand risk engine for electrical and electronic products, 2024

In the graph, the risk engine explicitly separates probability and consequence. The contribution of each P factor and T factor can easily be seen individually. It allows the influence of each P factor and T factor to be considered individually. It also allows products with very high technical safety hazard level or non-compliance probability factor to be considered on that factor alone.

New Zealand has carried out a number of comparison processes as the risk engine was being developed and are satisfied that the risk engine meets an acceptable level of accuracy, repeatability and reliability; however, it was noted that different societies place different tolerance levels on certain products, often reflecting the society's own safety awareness.

Challenges

Reviews of the risk engine and its factors have been carried out several times since it has been developed. Technologies may have advanced from time to time and the determined risk level for some products may have changed due to the traditional safety specifications for those products, and consumer behaviors for the way they use a particular product may also have changed.

As electrical energy takes over from liquid and gaseous fuels, the risk associated with the use of electrical energy will change, with the consequence that a reliable risk assessment tool will become necessary for regulators.

Basics of Risk Management in Trade

The Working Party on Regulatory Cooperation and Standardization Policies (WP.6) was founded in the 1970s as a forum for exchange on the harmonization of non-agricultural product regulations. WP.6 aims to promote regulatory cooperation, standardization policies and activities which contribute towards reducing technical barriers to trade, promoting sustainable development in all its dimensions including, for example, gender equality, climate and environmental protection, circular economy and the adaptation to new technologies.

WP.6 is the focal point within the UNECE on quality infrastructure. Quality infrastructure is comprised of regulations, structures, and bodies (such as accreditation, metrology, standards development bodies) that exist in a country/economy for supporting trade on a fair market to promote safe products and services in a sustainable society. The WP.6 subgroups allow stakeholders to work together to develop recommendations and exchange on best practices: the Advisory Group on Market Surveillance (MARS), the Group of Experts on Risk Management in Regulatory Systems (GRM), and the Ad-Hoc Team of Specialists on Standardization and Regulatory Techniques (START).

This work is hosted under the UNECE Market Access Section within the UNECE Economic Cooperation and Trade Division (ECTD). The ECTD assists member States with economic integration and in promoting and enabling a better policy, financial and regulatory environment. These support an inclusive and sustainable post-COVID-19 recovery and a transition to a more circular economy, e.g. through the promotion of gender-responsive standards and through digital and green transformations.

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Working Party on Regulatory Cooperation and Standardization Policies (WP.6)

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Website: http://www.unece.org

ISBN 978-92-1-003188-2

Designed and printed at United Nations, Geneva – 2416756 **(E)** – October 2024 – 279 – **ECE/TRADE/485**